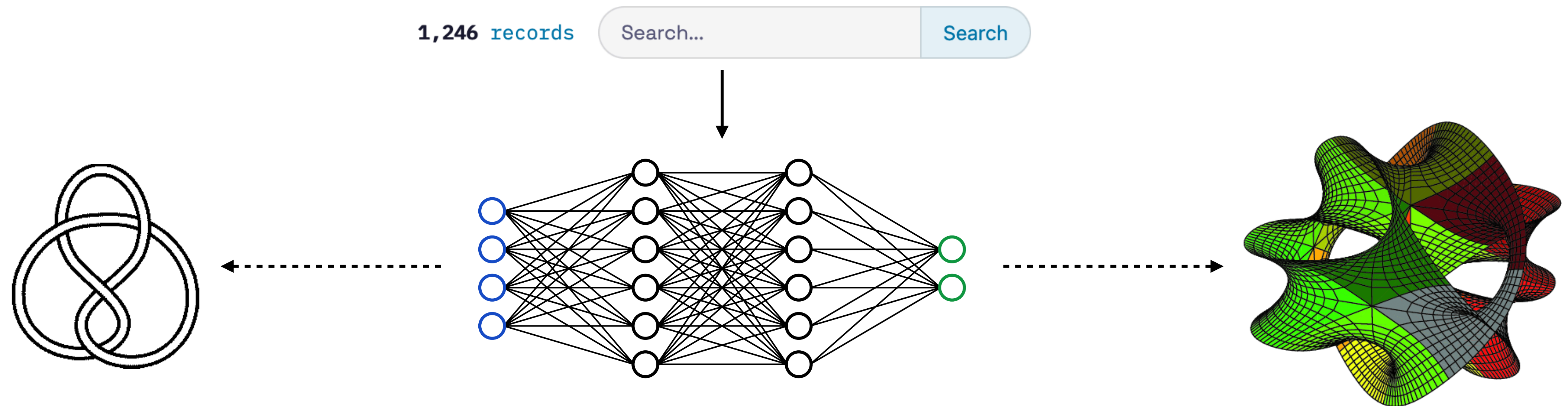


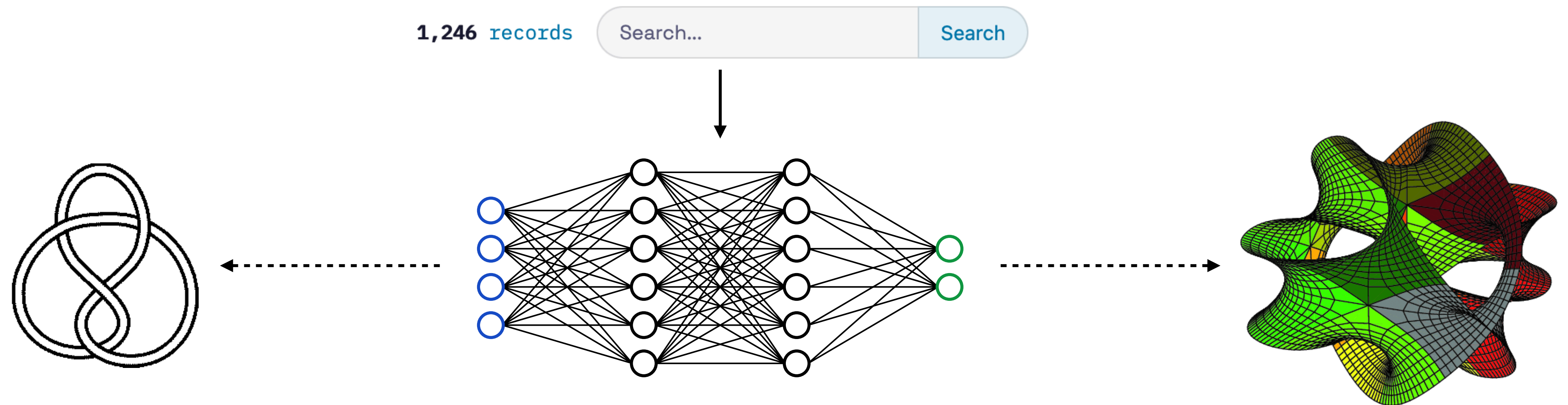
The first large-scale computation of the Gukov—Manolescu series



String Data 2025

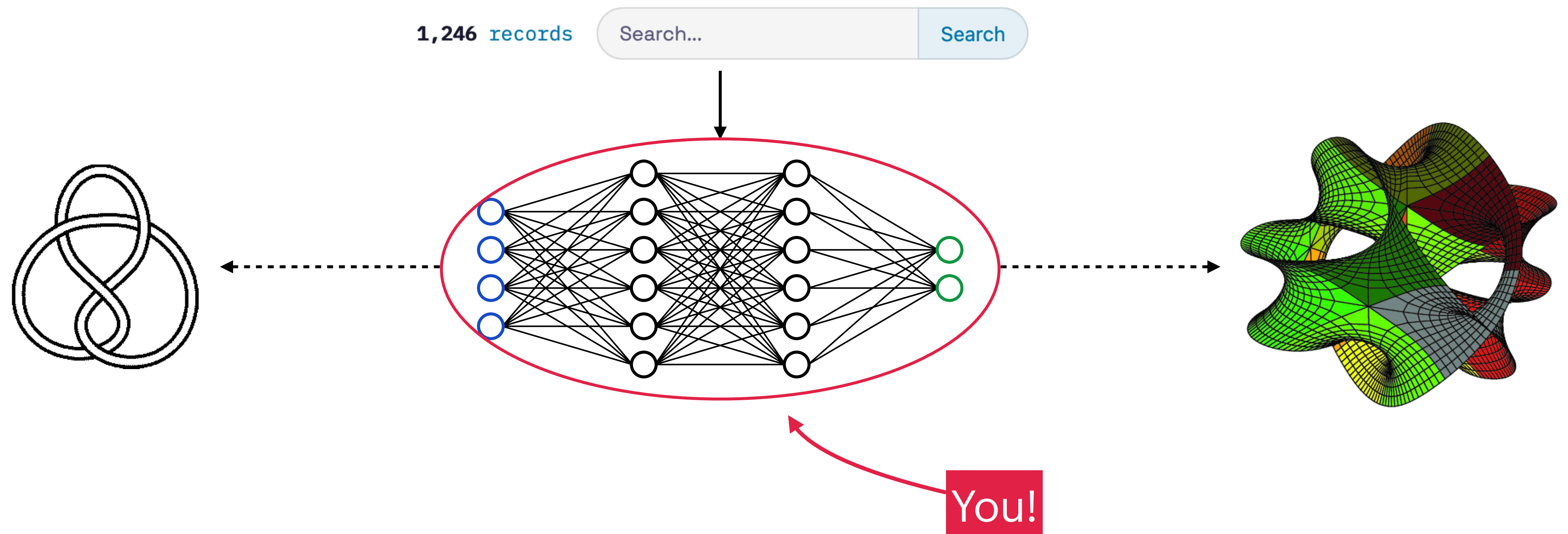
Lara San Martín Suárez (Caltech)

The first large-scale computation of the Gukov—Manolescu series



String Data 2025

Lara San Martín Suárez (Caltech)



Advisor



S. Gukov



P. Orland



D. Passaro



T. Saunders-A'Court



J. Svoboda

Collaborators

Today I'll be explaining

- Review of quantum invariants and construction of the GM series.
- Algorithm implemented to construct the database.
- How you can access and re-compute our data.
- First results obtained through human learning.

1990's

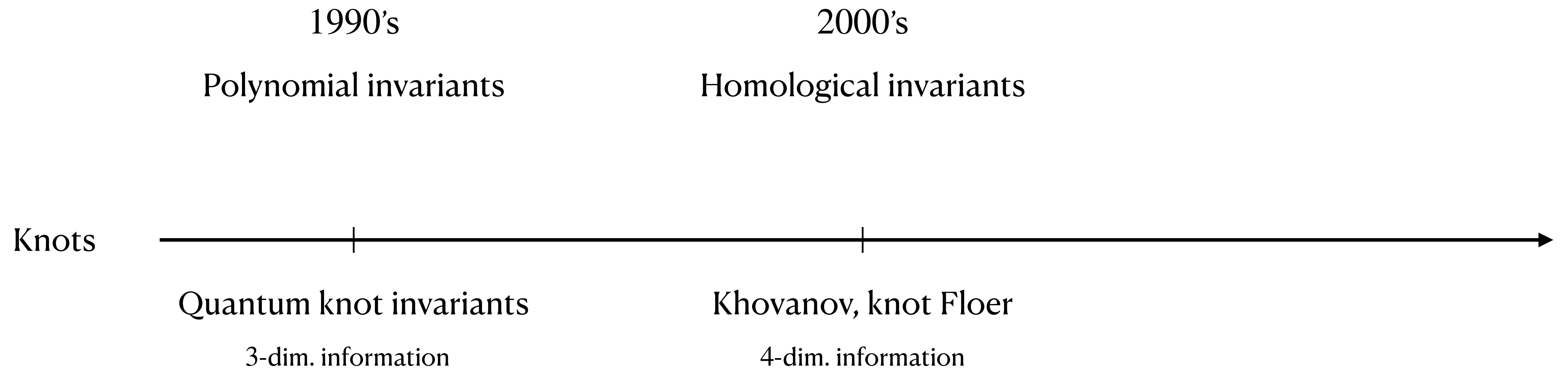
Polynomial invariants

Knots



Quantum knot invariants

3-dim. information



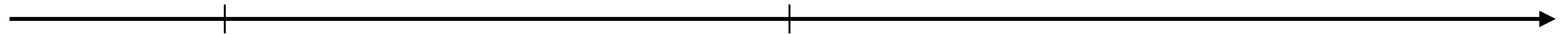
1990's

2000's

Polynomial invariants

Homological invariants

Knots



Quantum knot invariants

Khovanov, knot Floer

3-dim. information

4-dim. information

3-manifolds



WRT invariants

SU(2) Chern-Simons theory

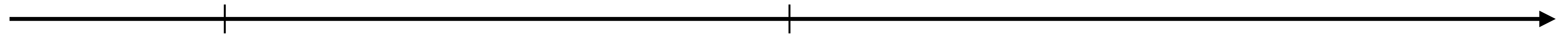
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WRT invariants

SU(2) Chern-Simons theory

Is there a homology theory for WRT?

1990's

2000's

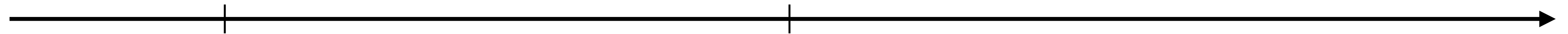
2015 – 2020's

Polynomial invariants

Homological invariants

Power series invariants

Knots



Quantum knot invariants

Khovanov, knot Floer

3-dim. information

4-dim. information

3-manifolds



WRT invariants

Is there a homology theory for WRT?

.....→ $\hat{Z}$ ⚠

SU(2) Chern-Simons theory

Counts of BPS states

Gukov-Pei-Putrov-Vafa

1990's

2000's

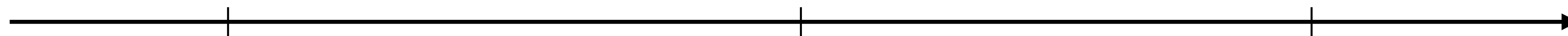
2015 – 2020's

Polynomial invariants

Homological invariants

Power series invariants

Knots



Quantum knot invariants

Khovanov, knot Floer

GM series (F_K)

3-dim. information

4-dim. information

3-manifolds



WRT invariants

Is there a homology theory for WRT?



$\hat{Z}$

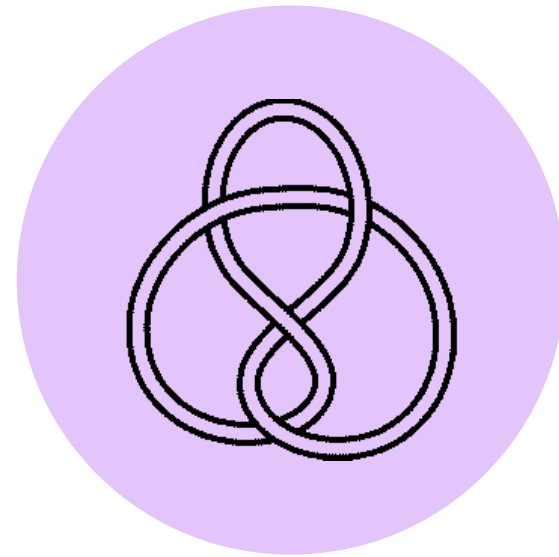


SU(2) Chern-Simons theory

Counts of BPS states

Gukov-Pei-Putrov-Vafa

Knot complement



$S^3 \setminus K$

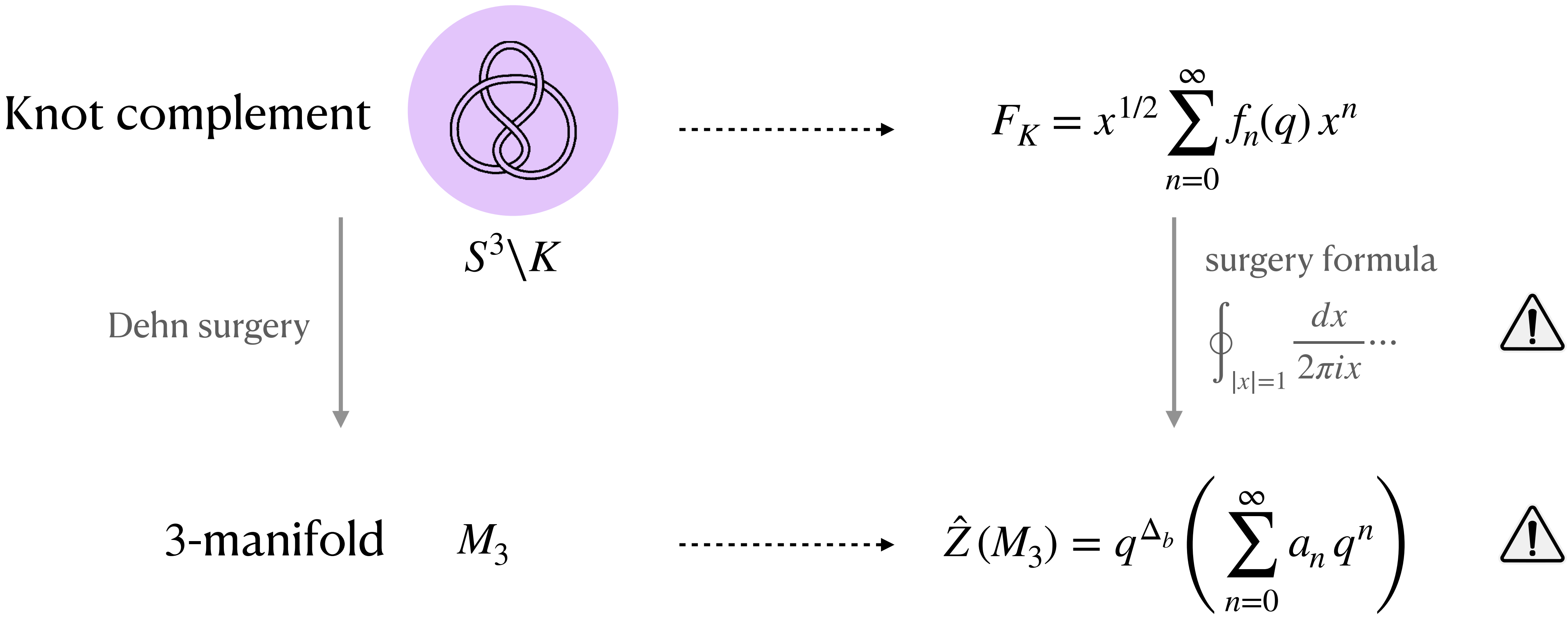
Dehn surgery

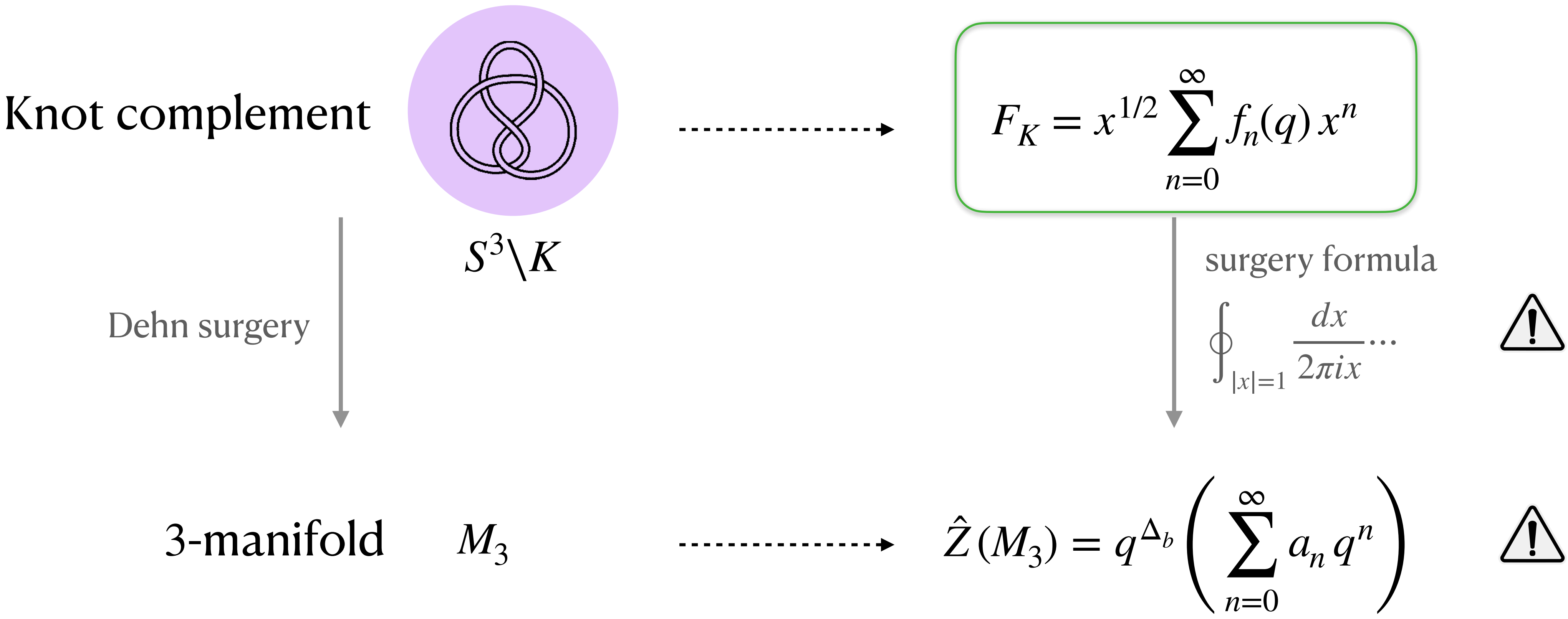


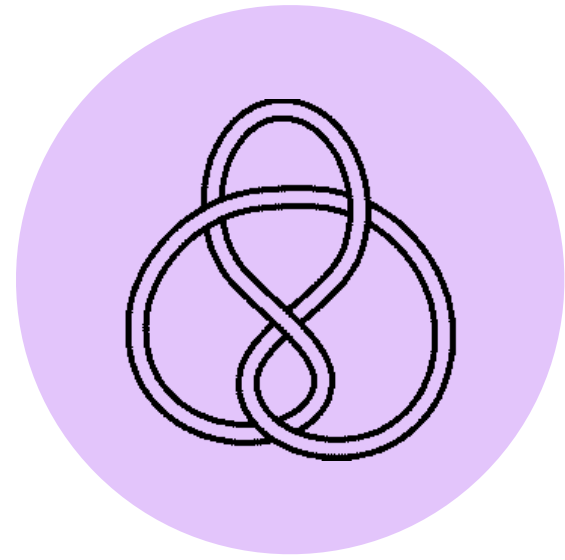
3-manifold M_3

Theorem (Lickorish, Wallace):

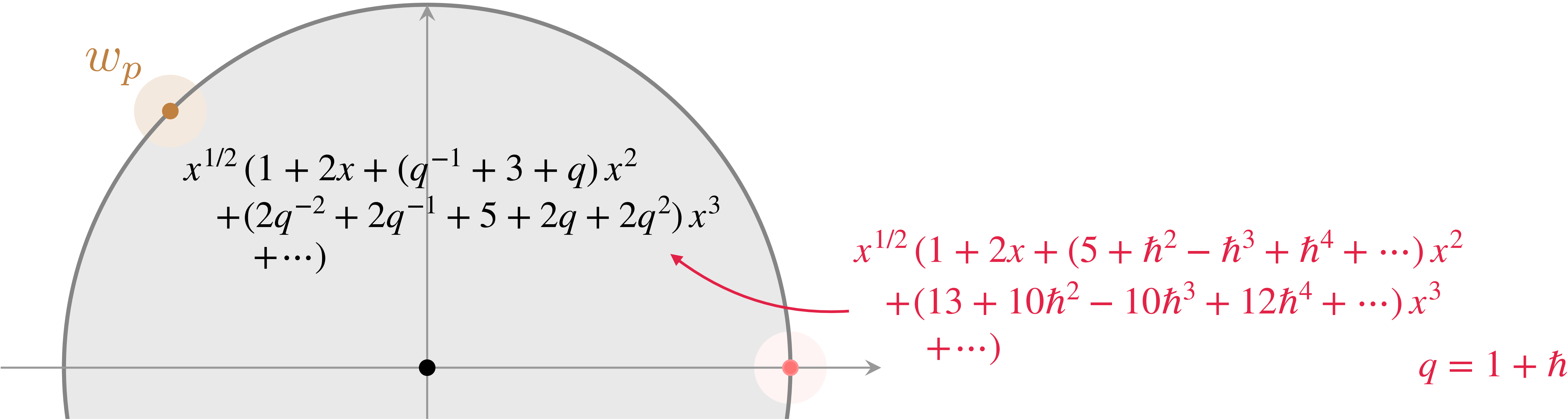
Every connected, oriented, closed
3-manifold arises by performing Dehn
surgery along a link $K \subset S^3$.





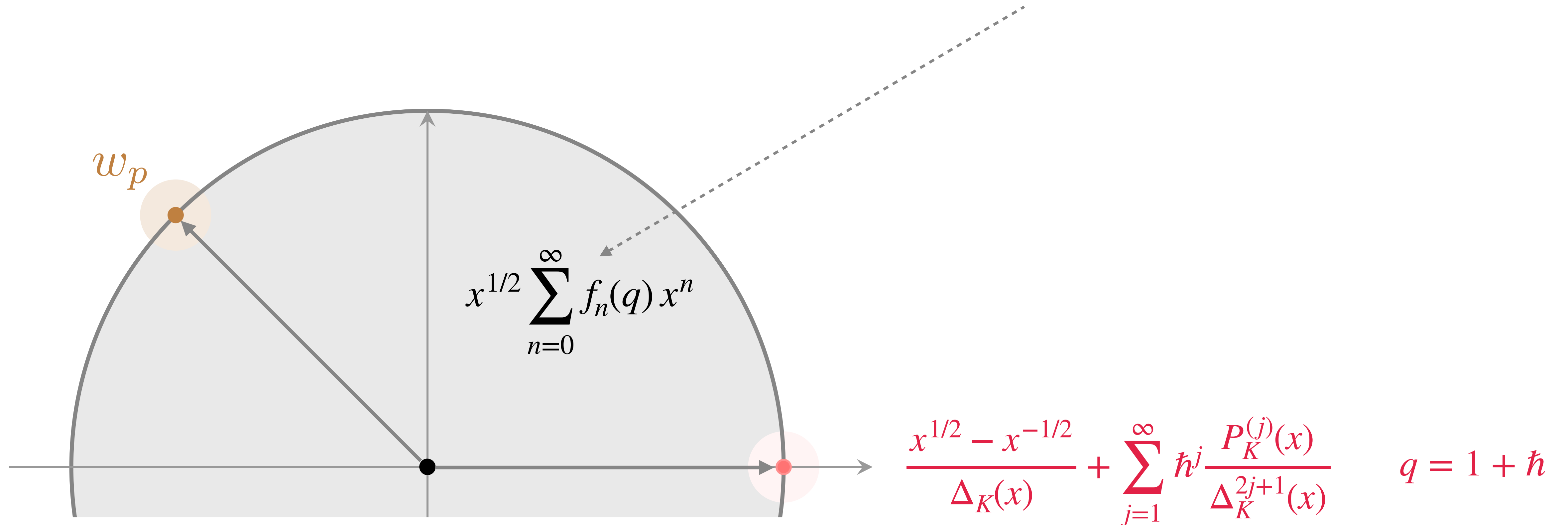


Gukov-Manolescu, 2018: resurgent analysis at $q = 1$



Gukov-Manolescu, 2018: resurgent analysis at $q = 1$

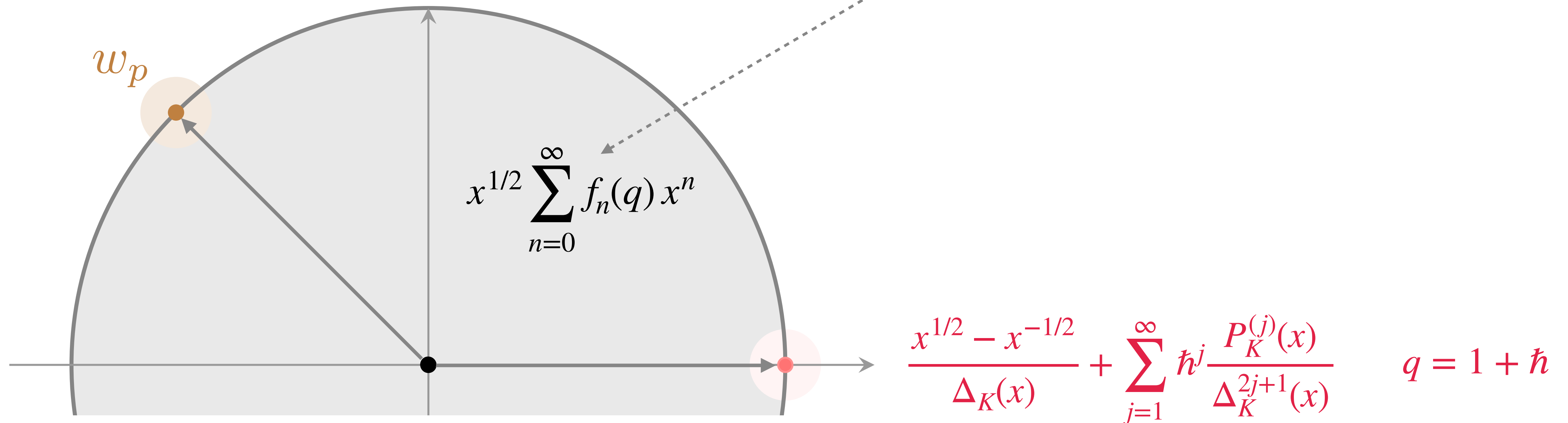
S. Park, 2020–21: quantum groups at generic q



Alexander polynomial: $\Delta(\nearrow \searrow) - \Delta(\nwarrow \swarrow) = \left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right) \Delta(\searrow \swarrow) \quad \Delta(\bigcirc) = 1$

Gukov-Manolescu, 2018: resurgent analysis at $q = 1$

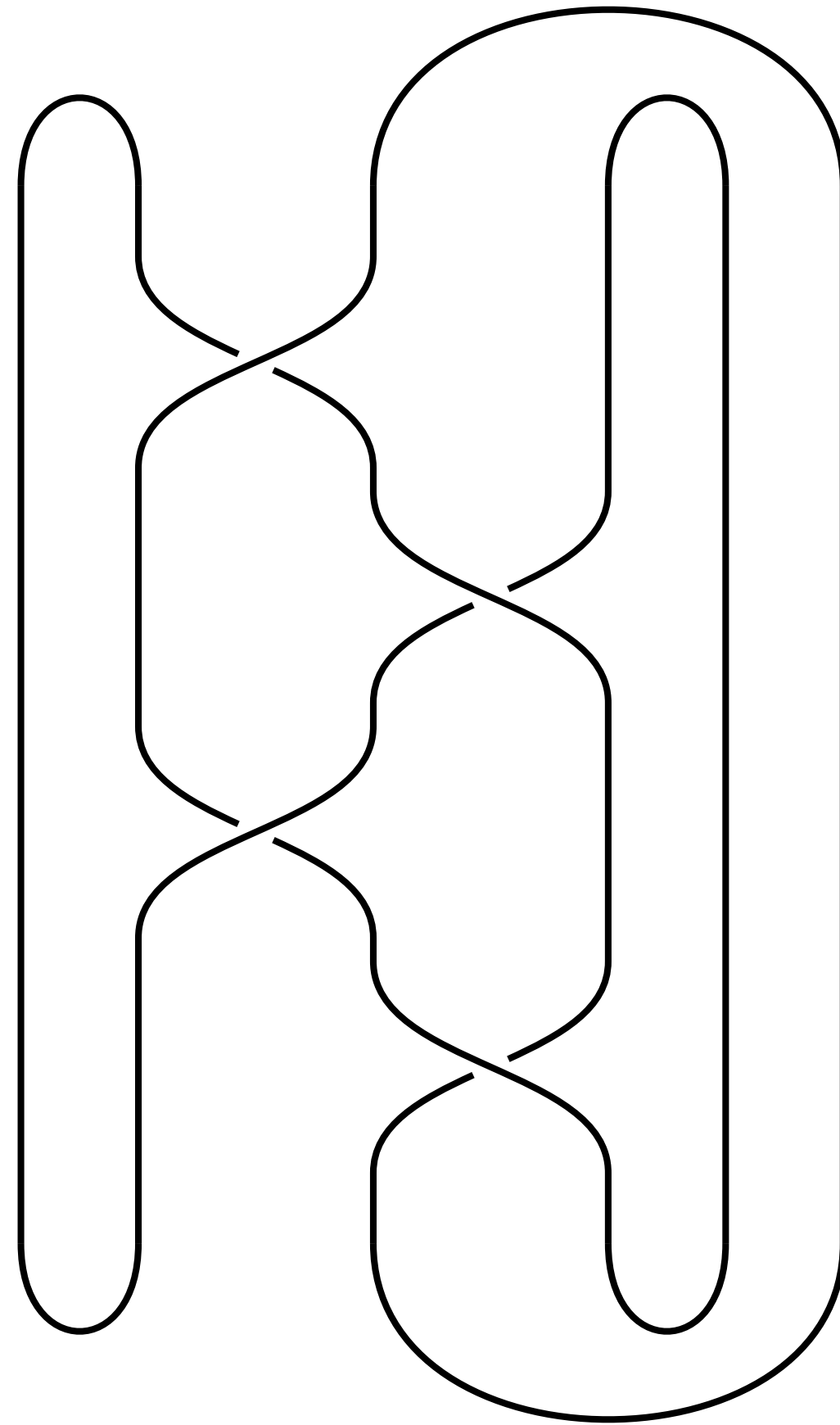
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Alexander polynomial: $\Delta(\nearrow \searrow) - \Delta(\nwarrow \swarrow) = \left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right) \Delta(\searrow \swarrow) \quad \Delta(\bigcirc) = 1$

Reshetikhin—Turaev: state sum over finite-dimensional modules of $U_q(\mathfrak{sl}_2)$ at generic q

1990's:

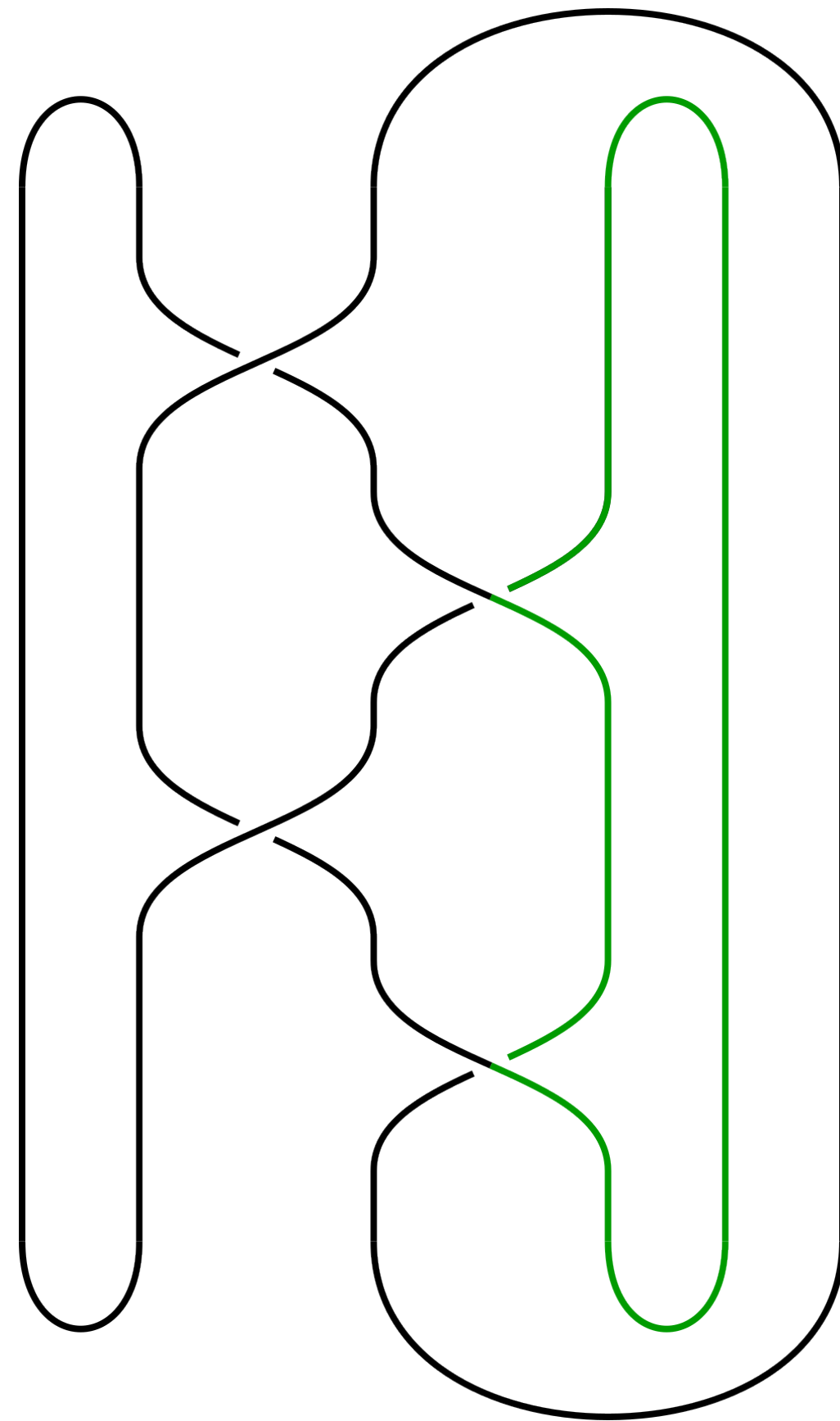


Quantum trace over a finite-dimensional vector space

$$V_n(-n) \underset{f}{\overset{e}{\rightleftharpoons}} V_n(-n+2) \underset{f}{\overset{e}{\rightleftharpoons}} \cdots \underset{f}{\overset{e}{\rightleftharpoons}} V_n(n-2) \underset{f}{\overset{e}{\rightleftharpoons}} V_n(n)$$

Park: state sum over (infinite-dimensional) Verma modules of $U_q(\mathfrak{sl}_2)$ at generic q

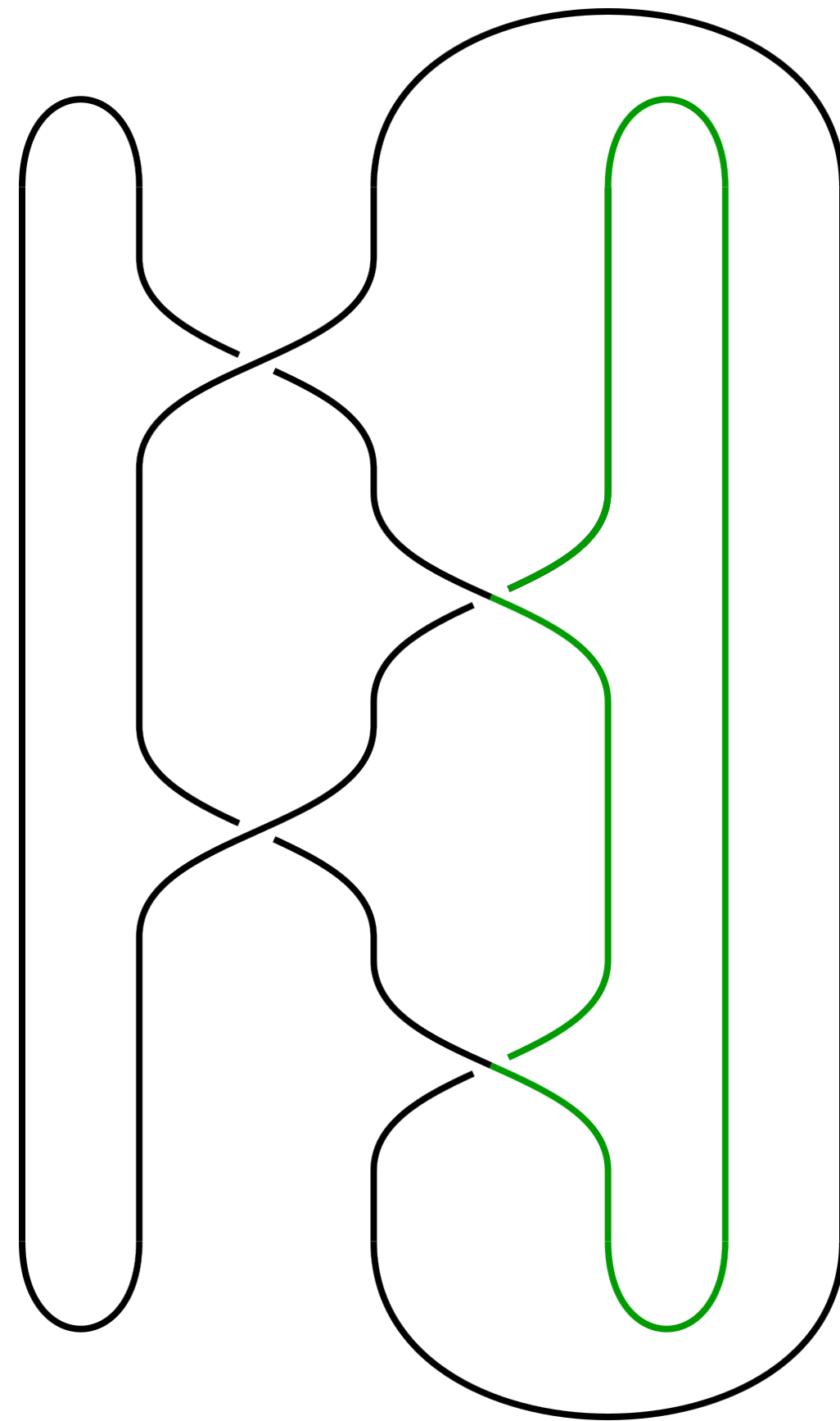
2020's:



Modified quantum trace over an
infinite-dimensional vector space
dependent on the **choice of a**
(simple) multicycle on the braid

Park: state sum over (infinite-dimensional) Verma modules of $U_q(\mathfrak{sl}_2)$ at generic q

2020's:

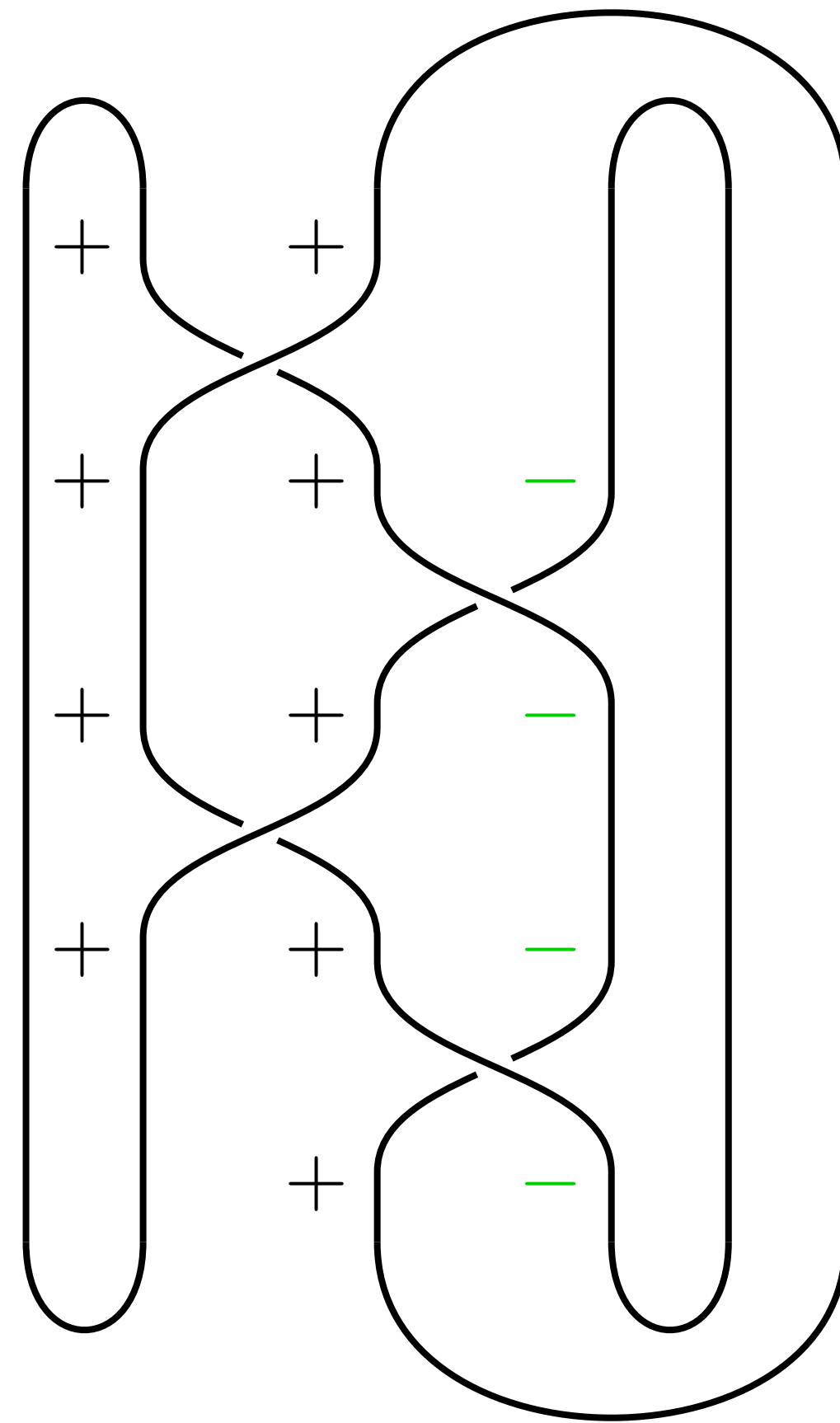


Modified quantum trace over an
infinite-dimensional vector space
dependent on the **choice of a**
(simple) multicycle on the braid

$$\cdots \xrightleftharpoons[f]{e} V_{\infty}(\lambda - 2) \xrightleftharpoons[f]{e} V_{\infty}(\lambda) \xrightleftharpoons[f]{e} V_{\infty}(\lambda + 2) \xrightleftharpoons[f]{e} V_{\infty}(\lambda + 4) \xrightleftharpoons[f]{e} \cdots$$

Park: state sum over (infinite-dimensional) Verma modules of $U_q(\mathfrak{sl}_2)$ at generic q

2020's:



Modified quantum trace over an
infinite-dimensional vector space
dependent on the **choice of a**
(simple) multicycle on the braid

Highest weight Verma

$$\cdots \xrightleftharpoons[f]{e} \langle v_1 \rangle \xrightleftharpoons[f]{e} \langle v_0 \rangle \xrightleftharpoons[f]{e} \langle v_{-1} \rangle \xrightleftharpoons[f]{e} \langle v_{-2} \rangle \xrightleftharpoons[f]{e} \cdots$$

Lowest weight Verma

1990's:

Braid $\rightsquigarrow$ (many) one-variable
polynomials.

No convergence issues.

2020's:

Braid & a choice of a **multicycle**
 $\rightsquigarrow$ a two-variable infinite series.

Convergence depends on the
choice of braid and multicycle.

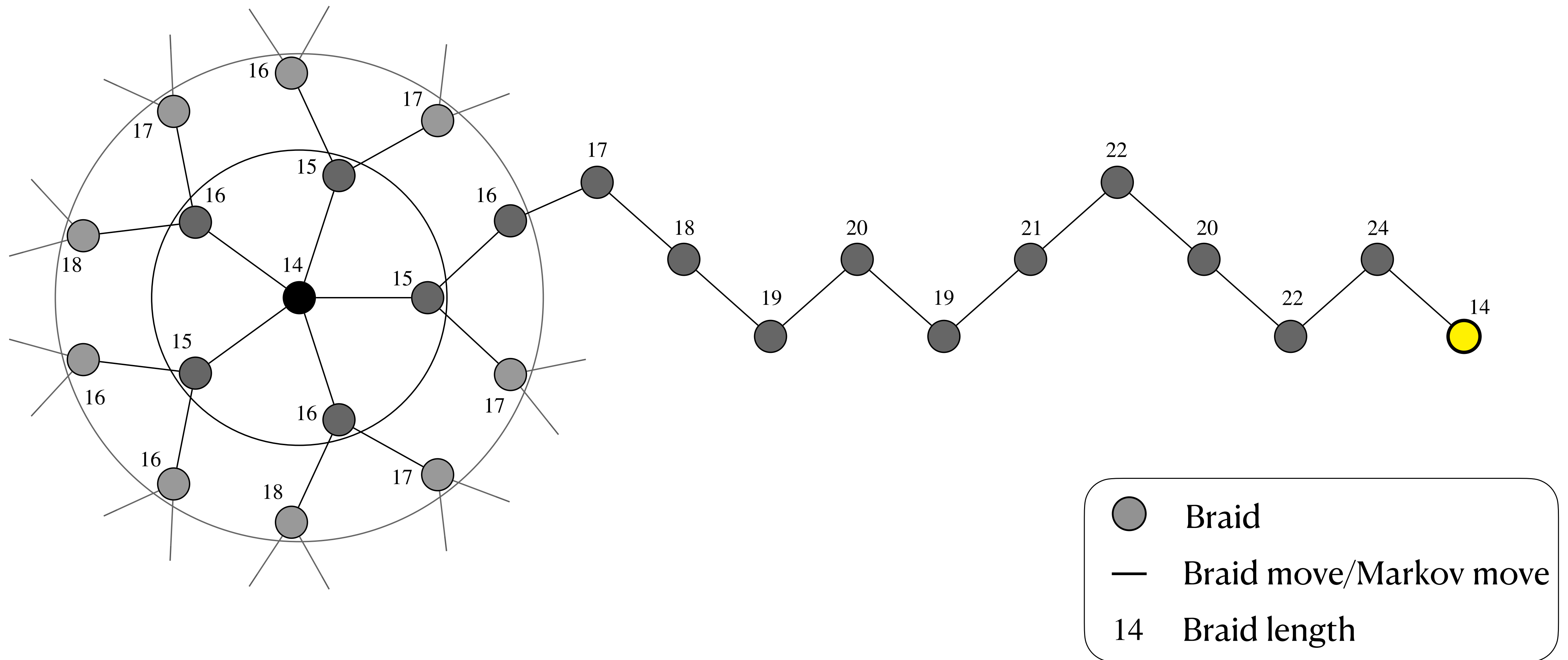
Why is this problem hard?

- We need a minimal length braid.

Why is this problem hard?

- We need a minimal length braid.
- Finding minimal length braids is hard.

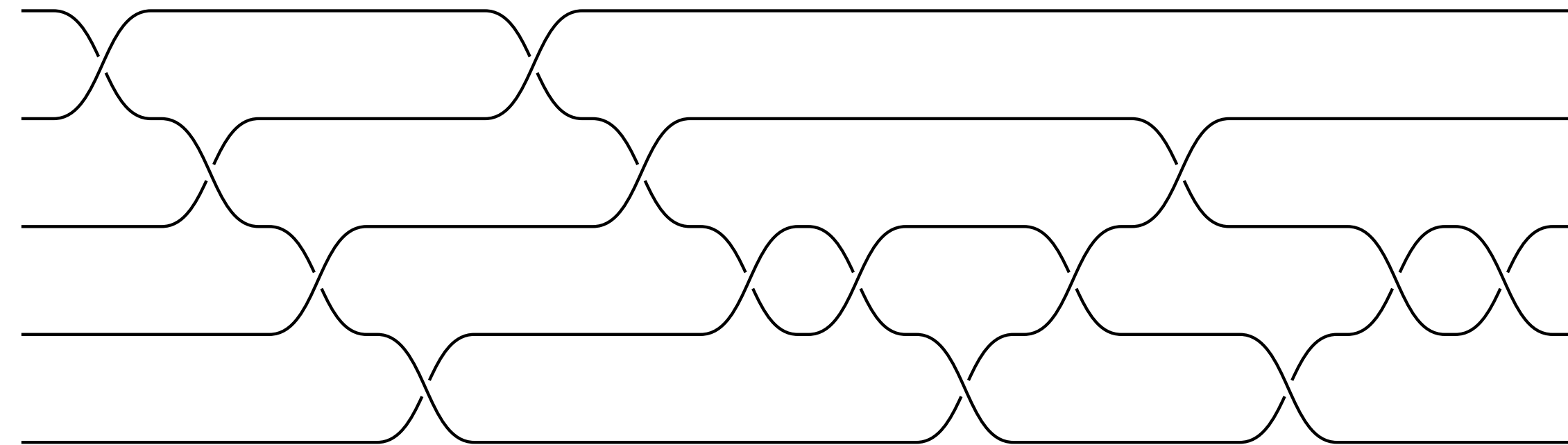
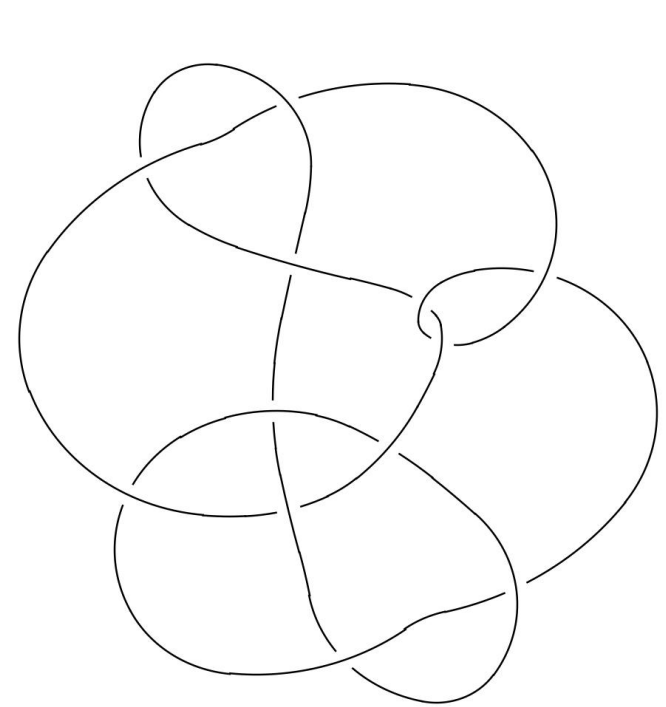
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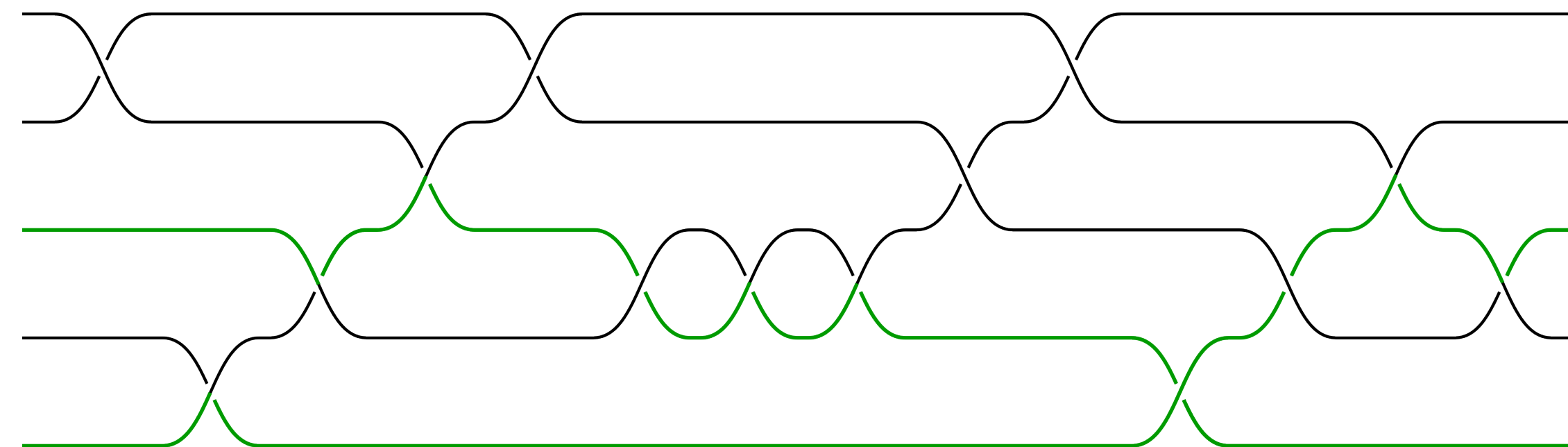
Why is this problem hard?

- We need a minimal length braid.
- Finding minimal length braids is hard.
- Finding a minimal length braid that admits a “convergent multicycle” is *very* hard.

Why is this problem hard?



$0/2^{14}$ multicycles
is “convergent”.



$1/2^{14}$ multicycles
is “convergent”.

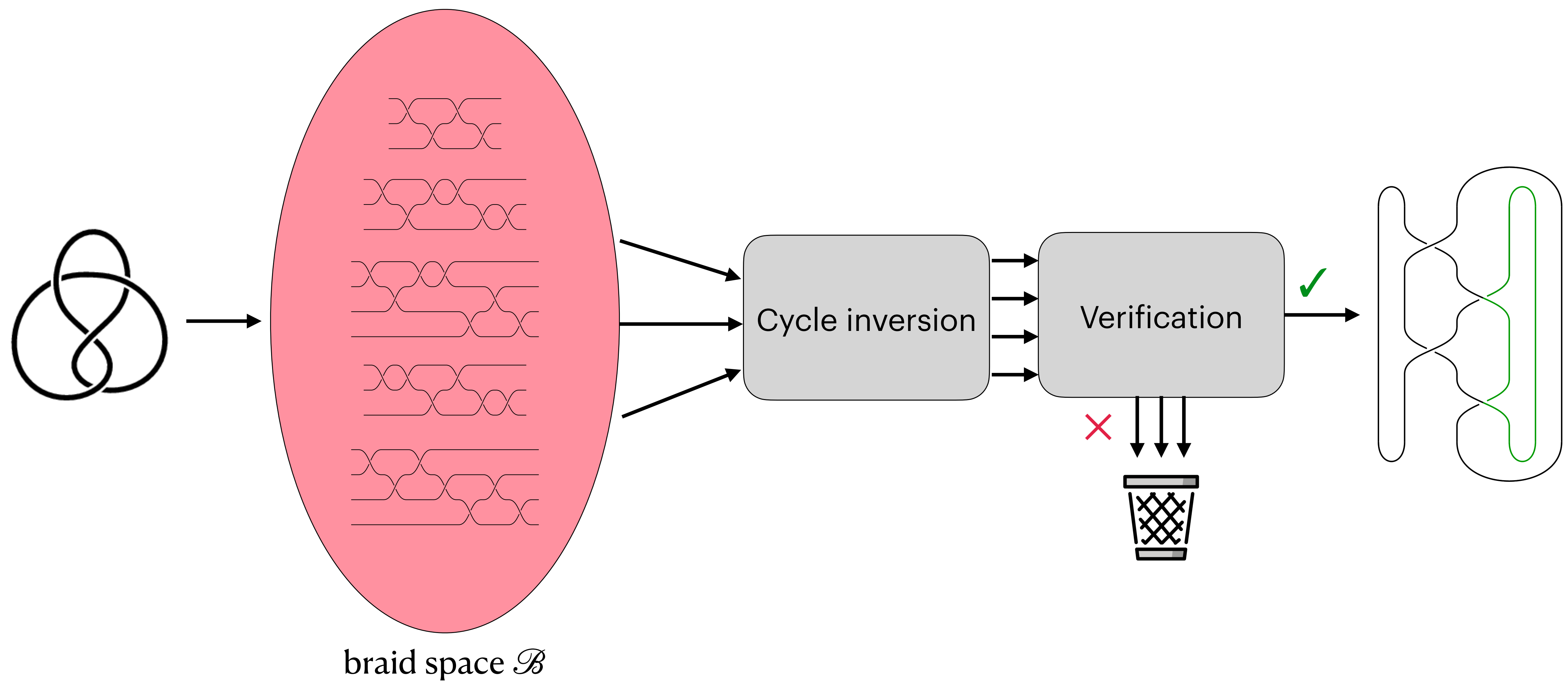
Why is this problem hard?

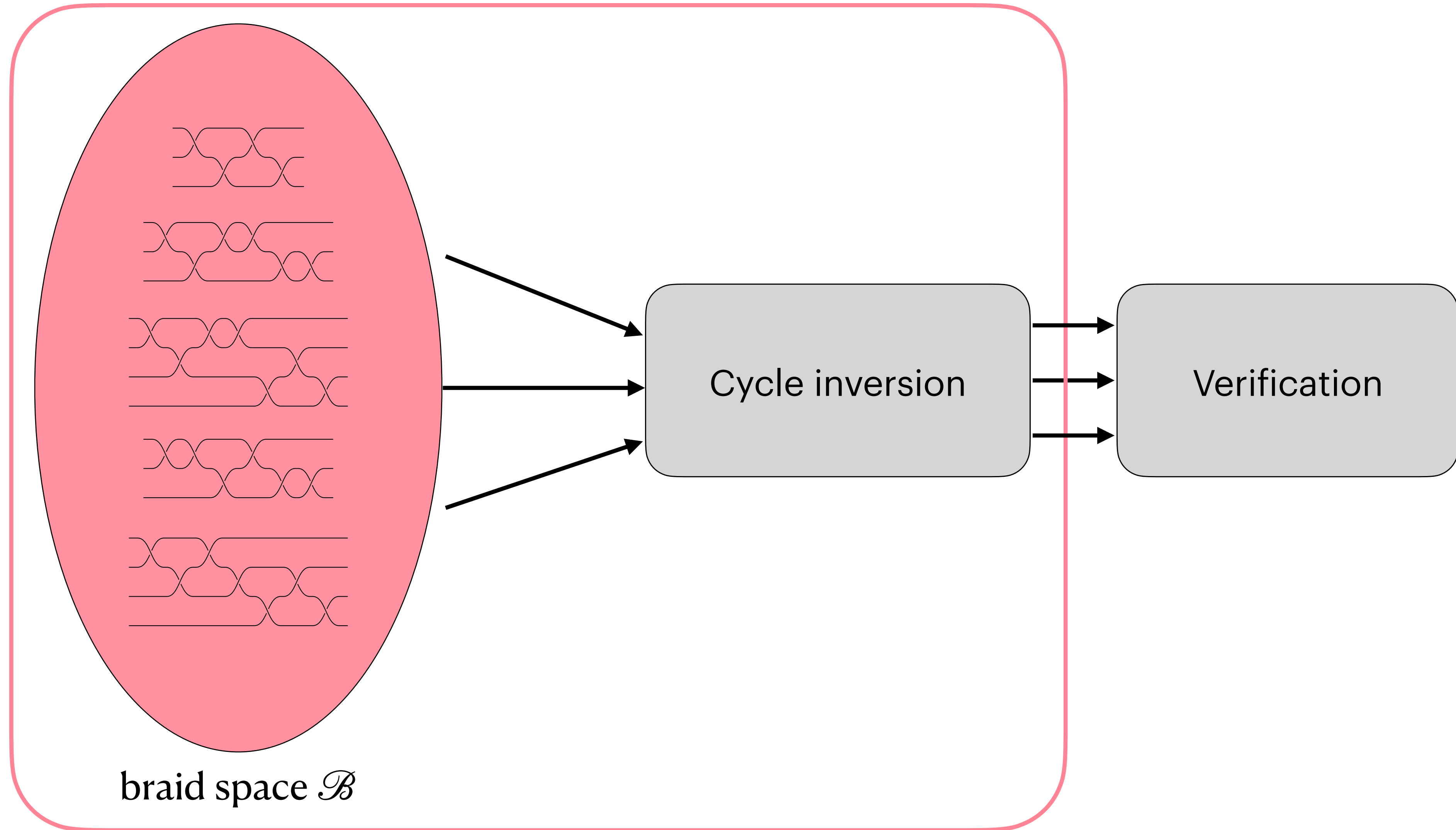
- We need a minimal length braid.
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- Finding a minimal length braid that admits a “convergent multicycle” is *very* hard.



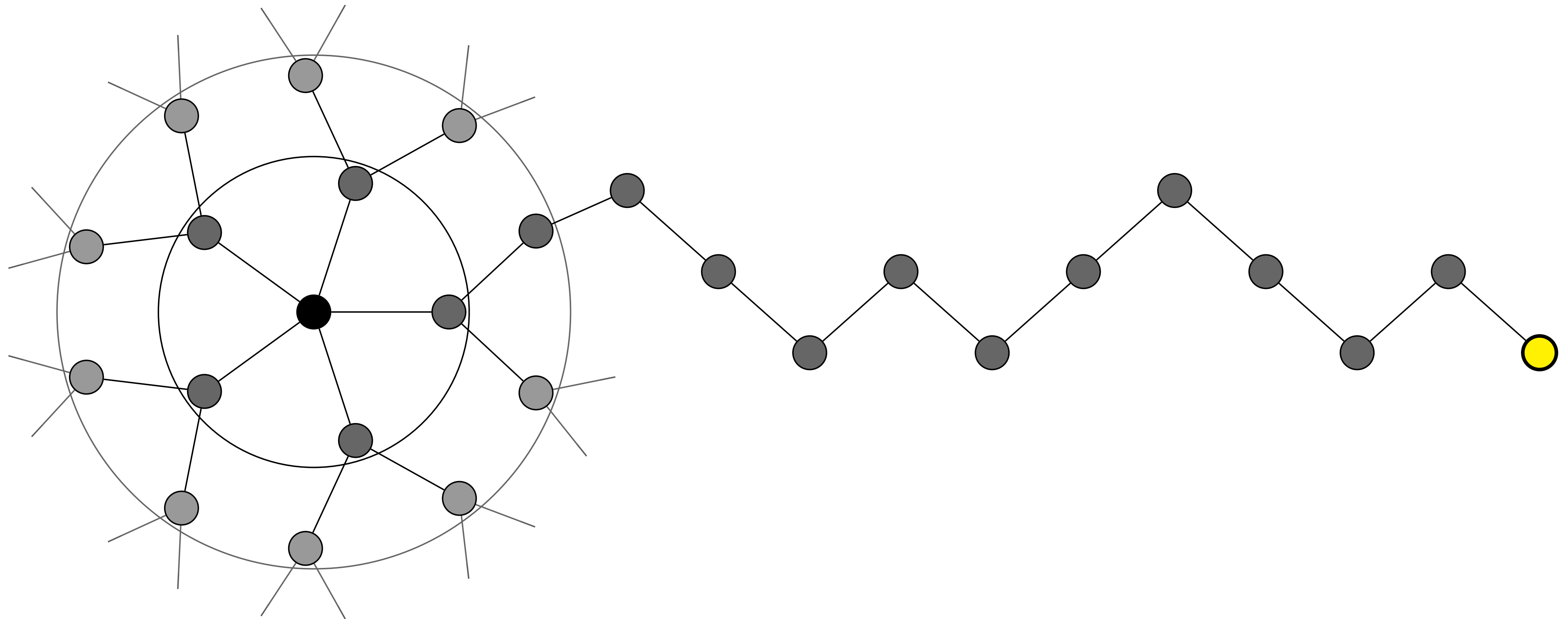
Side remark: we expect that this choice of multicycle has a geometrical meaning

So... how do we do it?



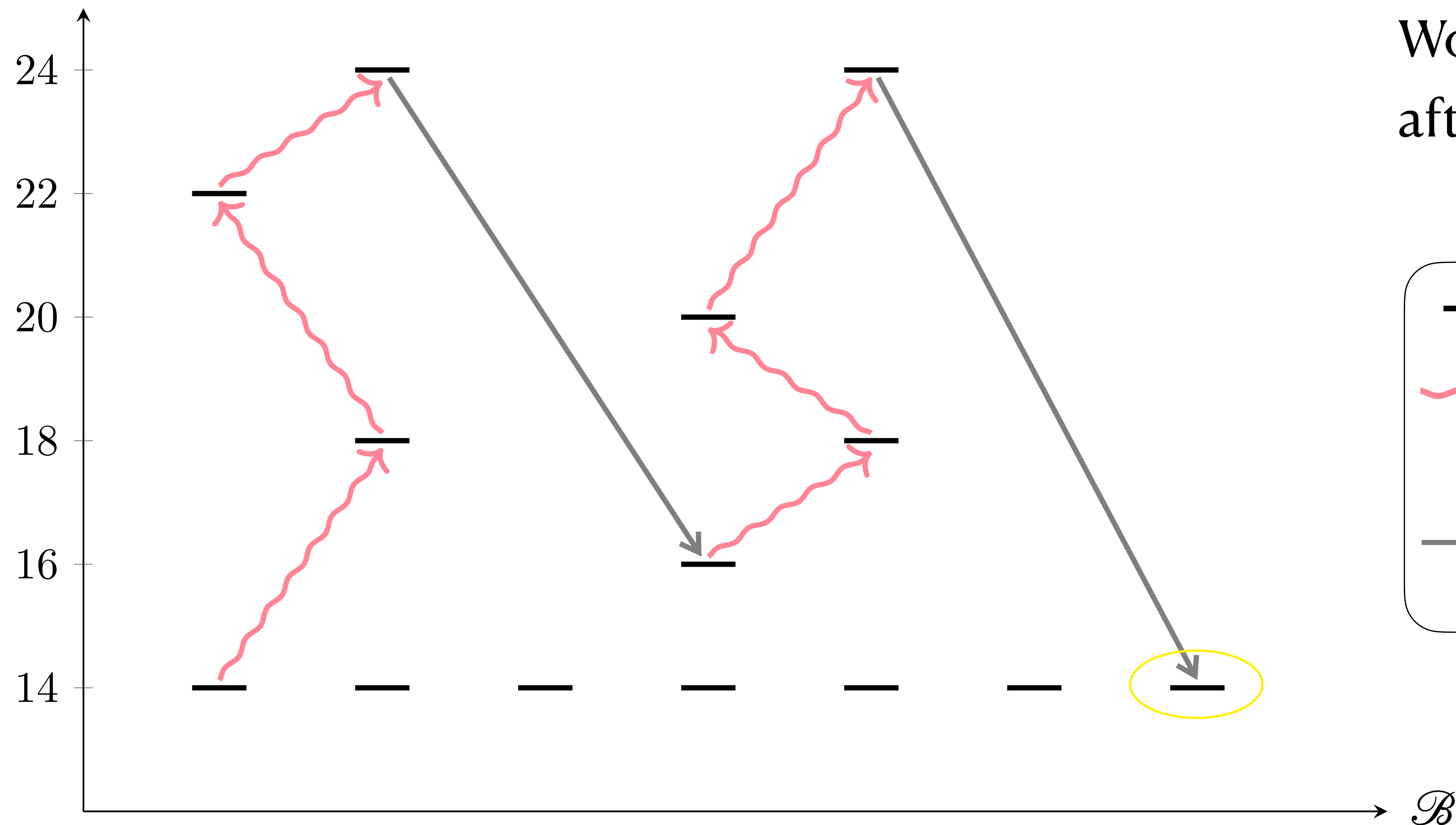


~~Search in $\mathcal{B}$~~



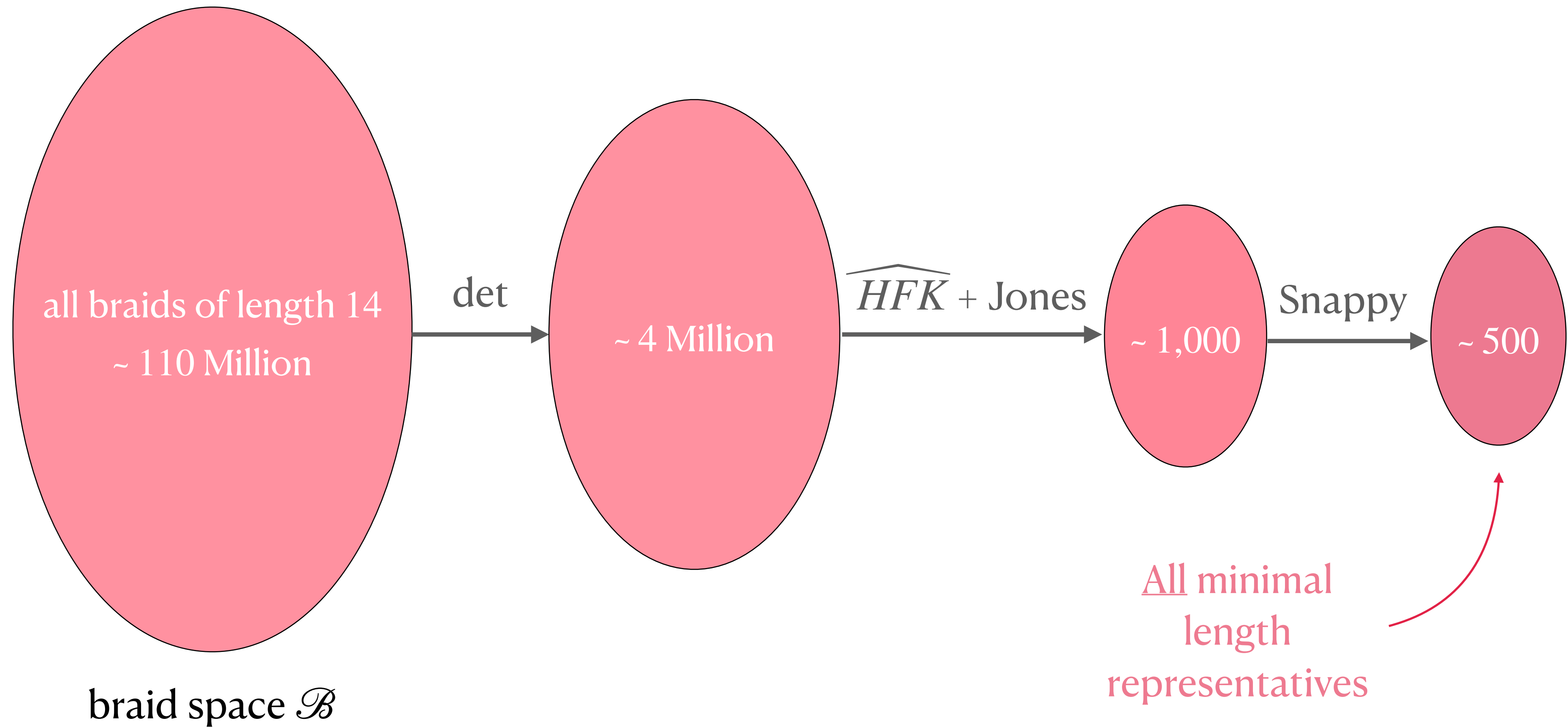
Random search in $\mathcal{B}$

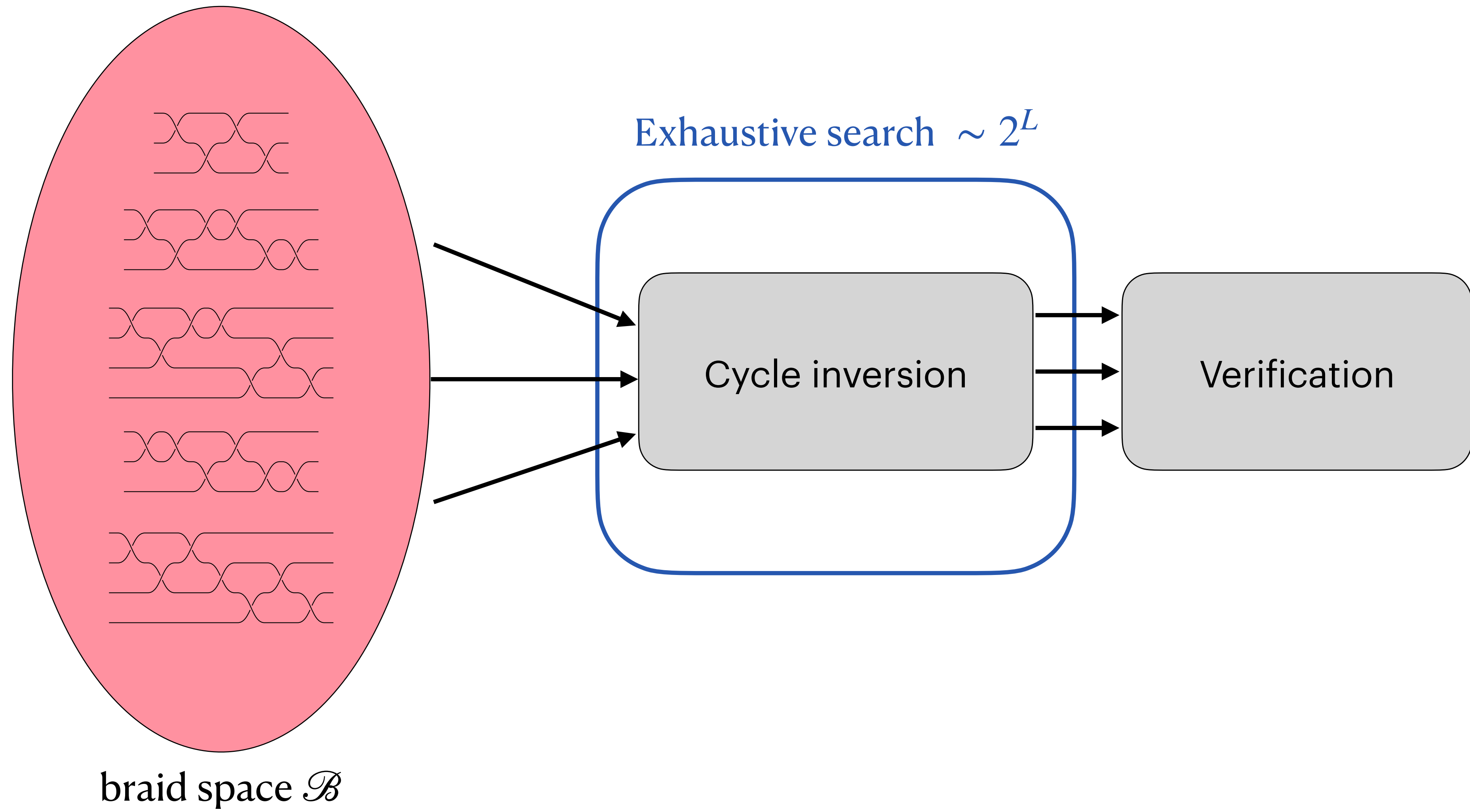
crossings

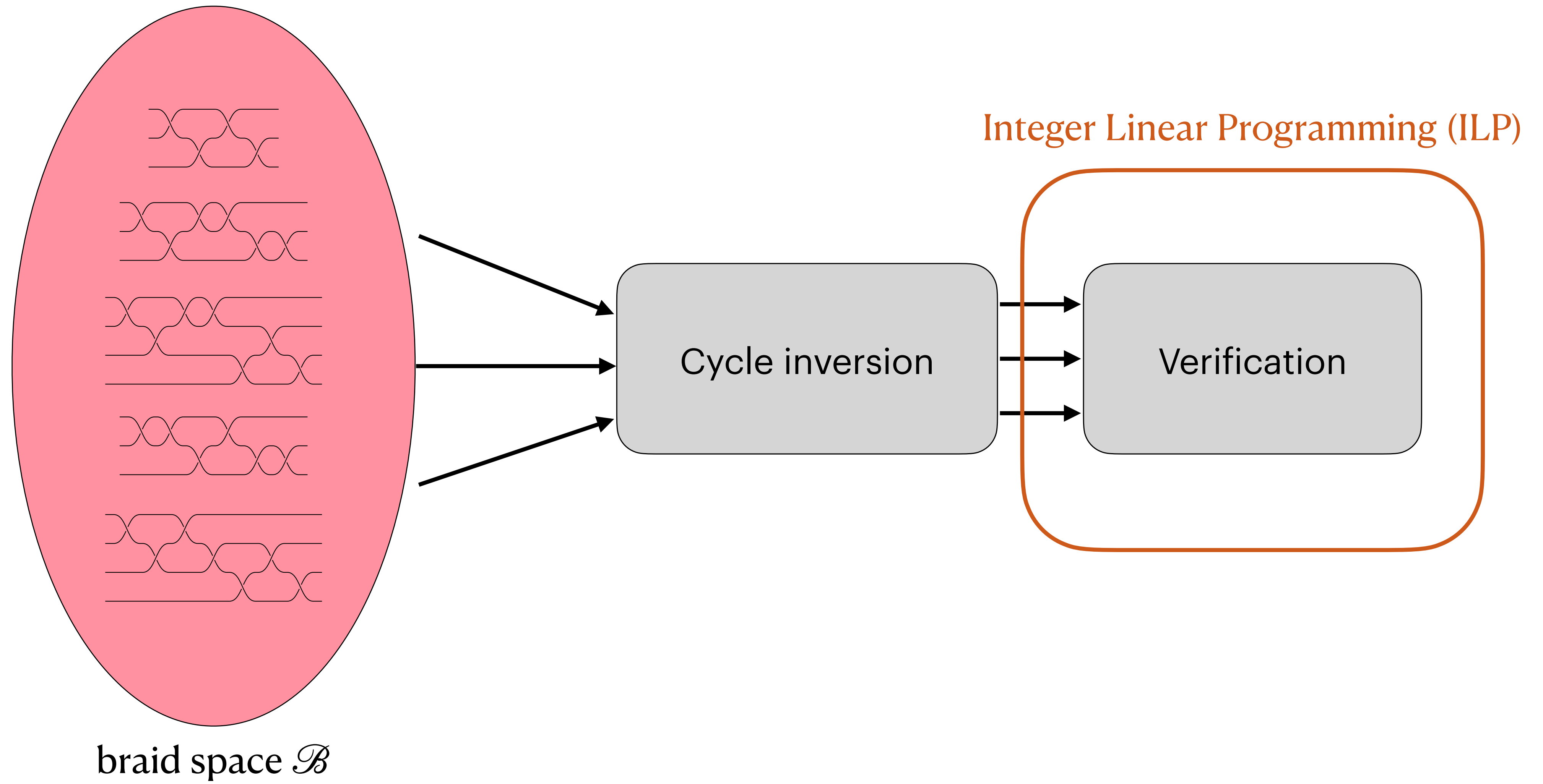


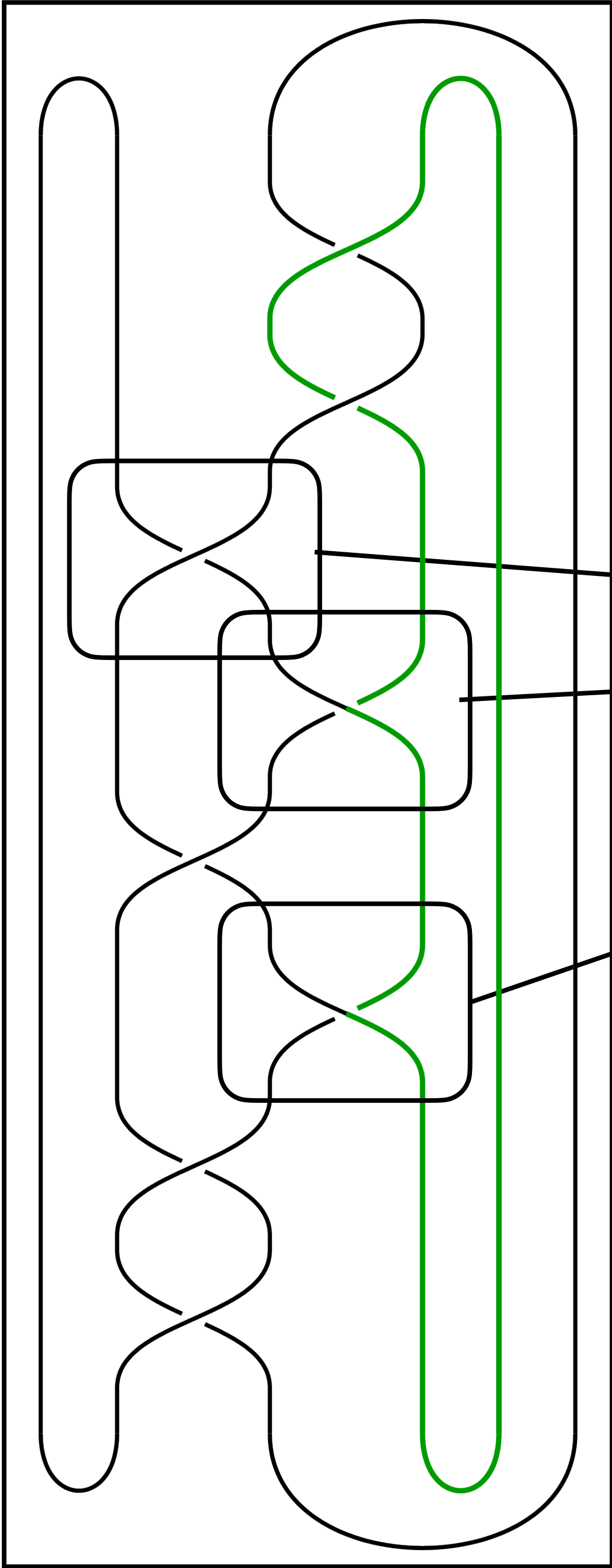
Works for **99.8%** of knots
after $\sim 10^6$ iterations.

For the remaining 0.2%:



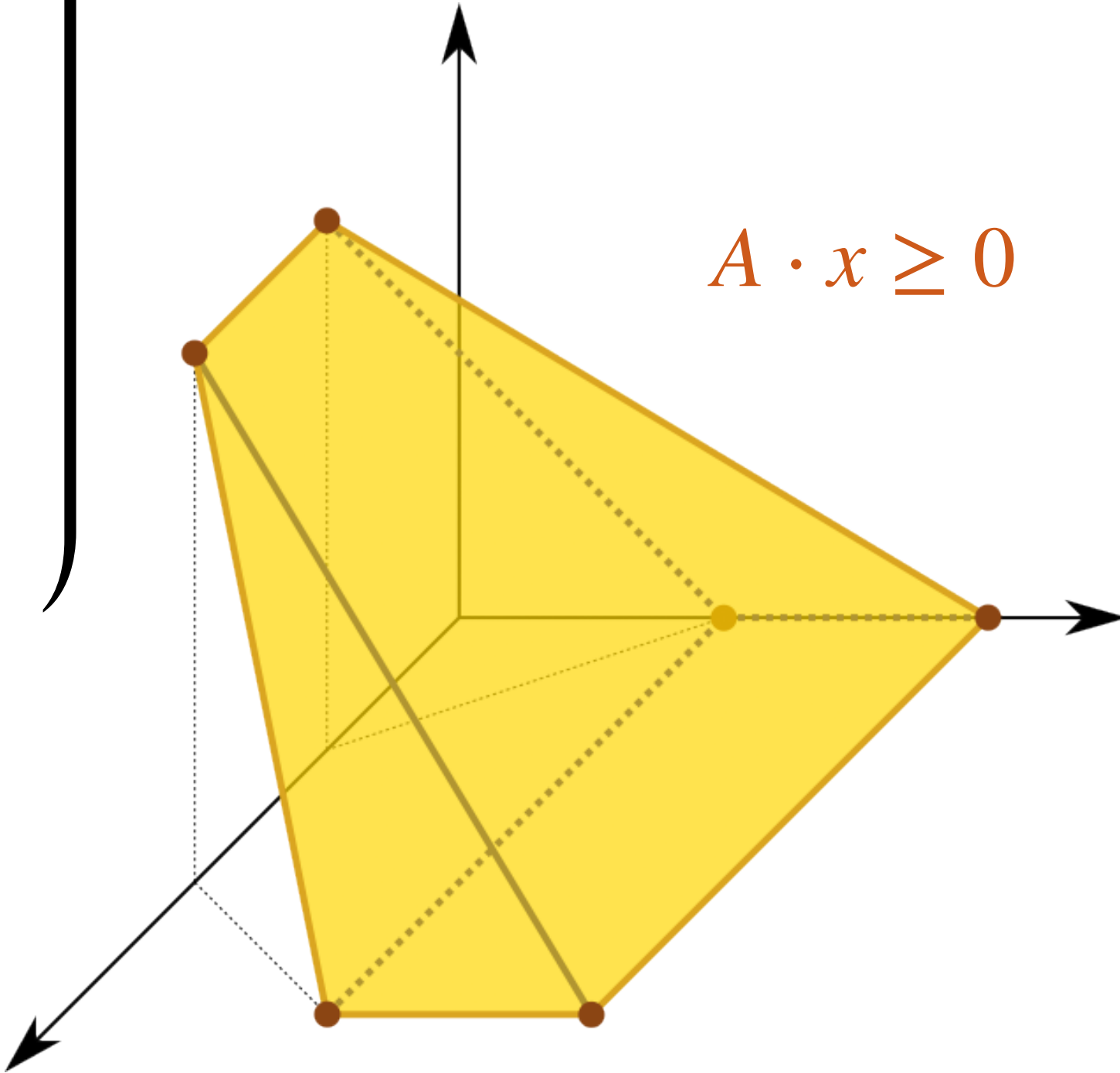




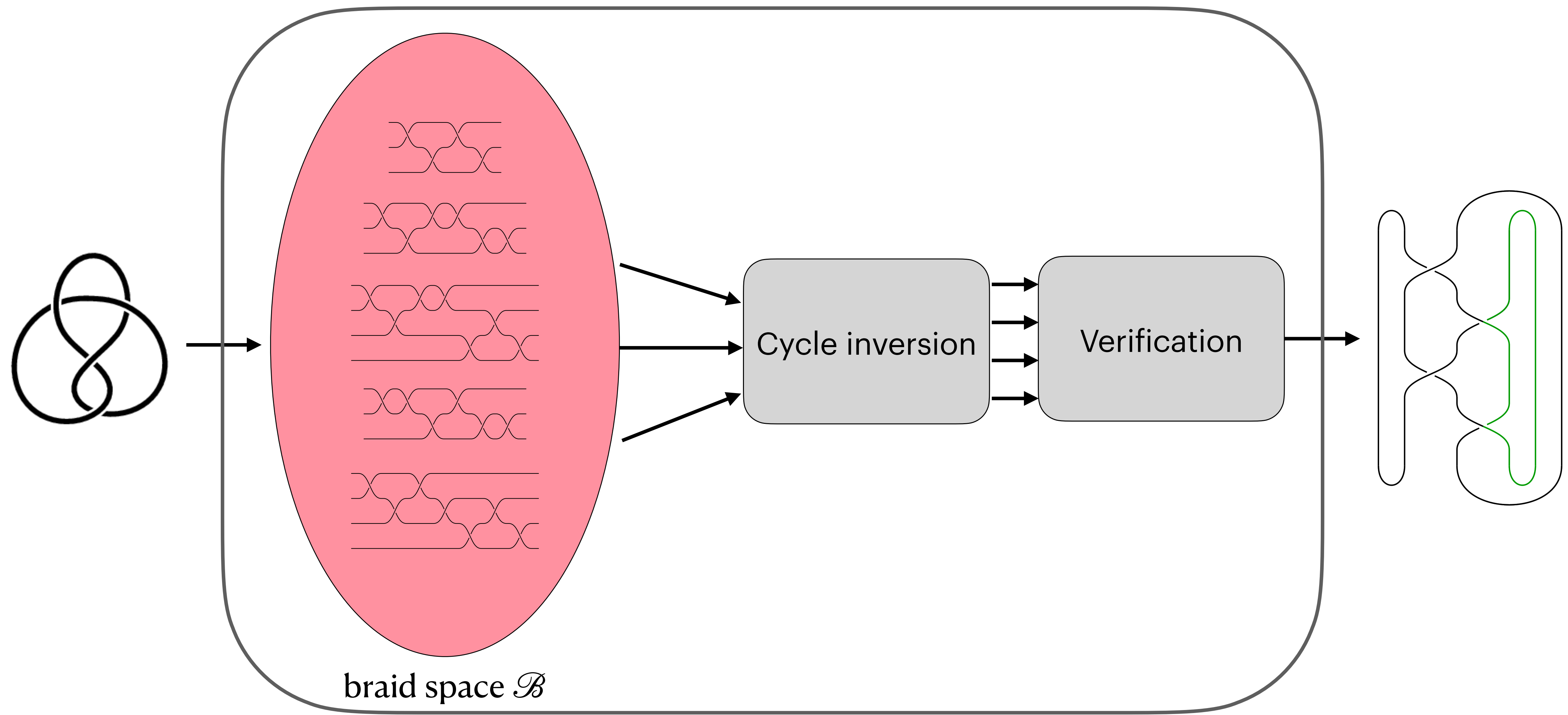


A

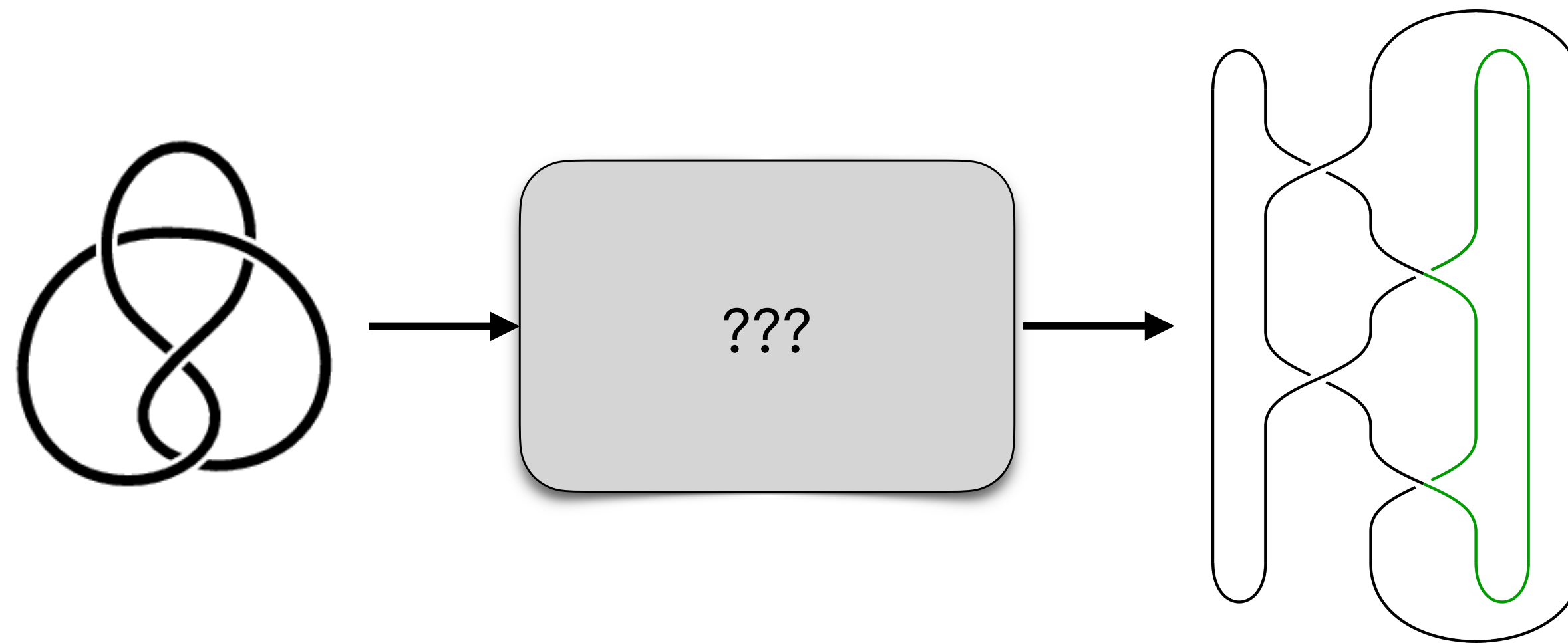
$$\begin{pmatrix} 3.5 & -1 & -0 & -1 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 \end{pmatrix}$$



If $A \cdot x \geq 0$, the resulting state sum is a Laurent power series in q, x



Can we *learn* this procedure?



The database

Result: 1,246 braids & choice of cycles

First large-scale computation of the F_K invariant

Home / Data / Gukov-Manolescu series (F_K)

Gukov-Manolescu series (F_K)

This table contains the the first few terms of the Gukov-Manolescu series under the column `fk_polynomial`. The braid and inversion datum used to compute the inverted state sum are listed under `braid` and `inversion_data`.

1,246 records

Search...

Search



CODE

CITE

↓ DOWNLOAD

PYTHON

MATHEMATICA

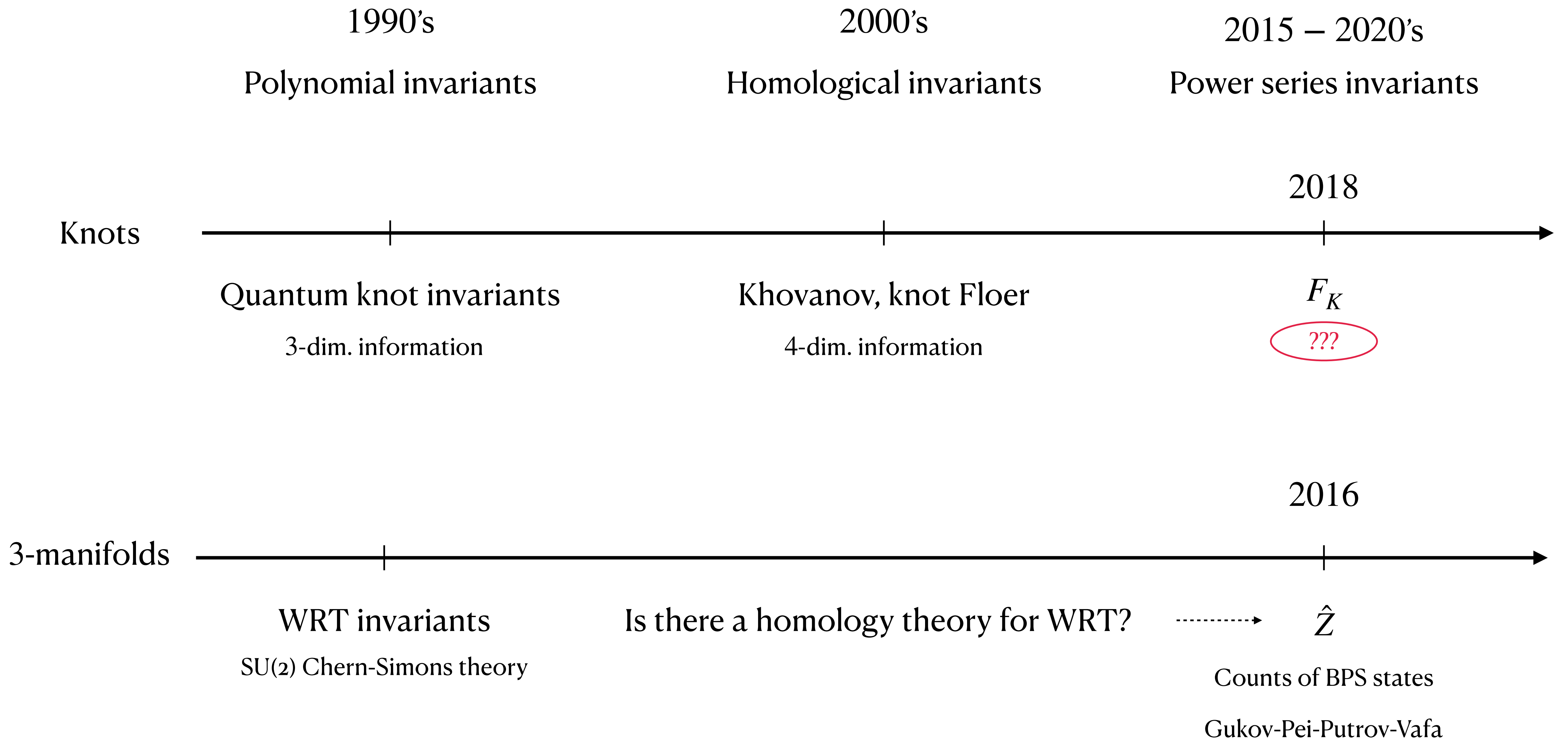
API

KNOTINFO_ID ?	BRAID ?	INVERSION_DATA ?	FK_POLYNOM
3_1	[1, 1, 1]	[-1, 1, -1, 1, -1, 1]	-q+q^2*x^2-
4_1	[1, -2, 1, -2]	[-1, 1, -1, 1, -1, 1, -1, 1]	-1-2*x-q^-:
5_1	[1, 1, 1, 1, 1]	[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]	-q^2*x+q^3:
6_2	[1, 1, 1, -2, 1, -2]	[-1, 1, -1, 1, -1, 1, -1, 1, -1, 1, -1, 1]	-q*x-2*q*x
6_3	[1, 1, -2, 1, -2, -2]	[1, 1, 1, 1, -1, 1, 1, 1, -1, 1, -1, 1]	-x-2*x^2+q
...	...	...	...



<https://topology.fyi>

First results



Theorem (Orland—Saunders-A’Court—S.—Svoboda): For a “nice” knot K ,

$$F_K(x, q) = x^{-1/2}(q^{\ell_K} x^{d_K} + \text{higher order terms in } x)$$

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If K is a homogeneous braid knot, or a **fibred knot of at most to 12-crossings**,

$$d_K = \max\{a \mid \widehat{HFK}(K, a) \neq 0\} = g_K$$

$$\ell_K = g_K - \min\{m \mid \widehat{HFK}_m(K, g_K) \neq 0\}$$

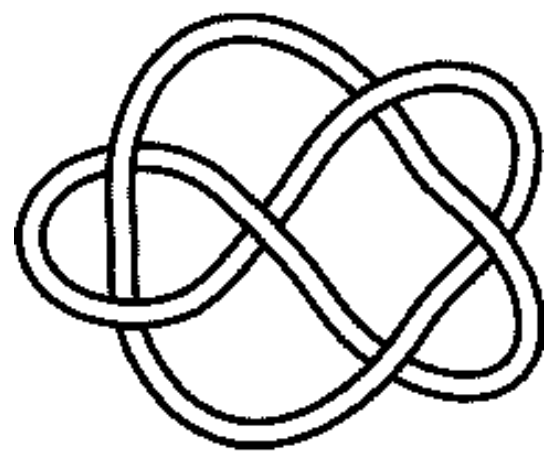
Theorem (Orland—Saunders-A’Court—S.—Svoboda): For a “nice” knot K ,

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$$\widehat{HFK}(6_2) = m a^2 + 3a + 3m^{-1} + 3m^{-2}a^{-1} + m^{-3}a^{-2}$$

$$F_{6_2}(x, q) = x^{-1/2}(q x^2 + 2q x^3 + (1 + 3q - q^2) x^4 + (2q^{-1} + 2 + 4q - 2q^2) x^5 + \cdots)$$

Corollary (Orland—Saunders-A’Court—S.—Svoboda): For a “nice” knot K ,

$$\ell_K = \sum_{i=0}^{2g_K-1} \frac{1}{i!} \frac{d^i \Delta_K^3}{dy^i} \bigg|_{y=0} \frac{dh_{2g_K-i}^K}{dq} \bigg|_{q=1}$$

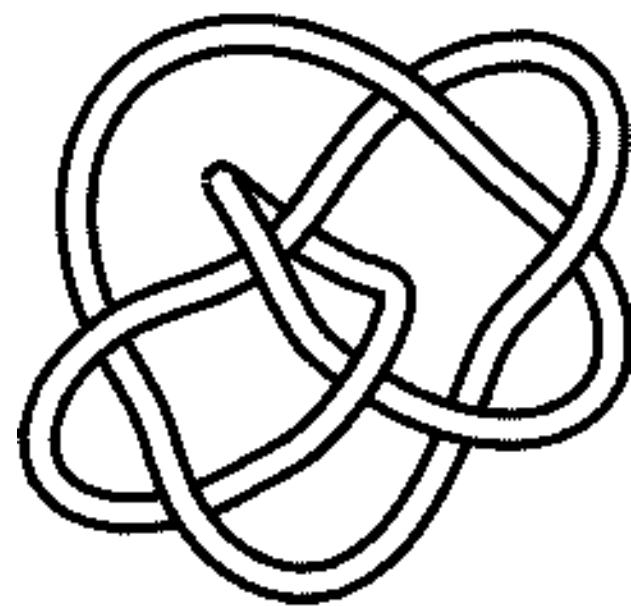
↖ Habiro coefficients
(Change of basis of Jones polynomials)

(Change of basis of Jones polynomials)
Habiro coefficients

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Homological grading



$$\begin{aligned} 4 &= 3 - (2030 - 8505 + 14076 - 11508 + 4662 - 756) \\ &= 3 - (-1) \end{aligned}$$

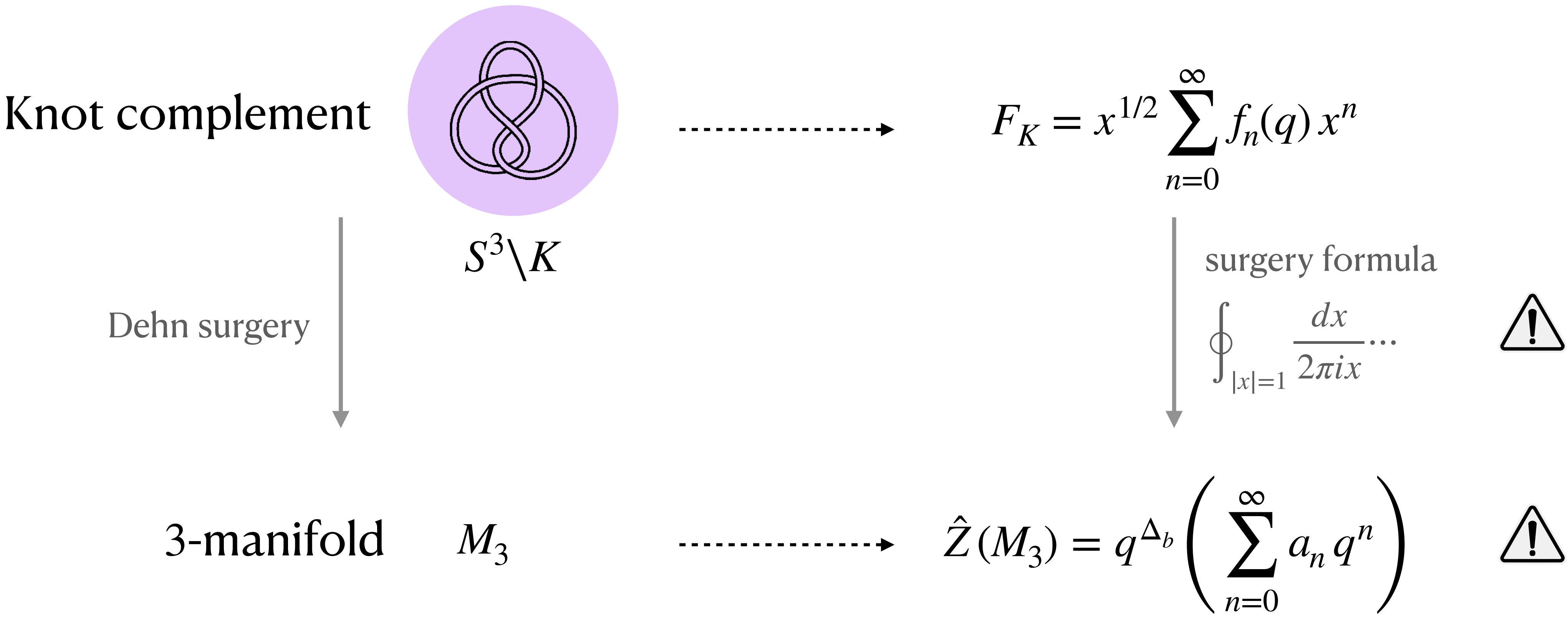
$$\text{Let } F_K = \sum_{n=0}^{\infty} f_n(q) x^n \text{ and } \delta_n(K) := \min \deg_q f_n(q)$$

Conjecture (Orland—Saunders-A’Court—S.—Svoboda): For a “nice” knot K ,

$$\delta_n(K) = An^2 + B(n)n + C(n)$$

where A is the inverse of a boundary slope in $S^3 \setminus K$.

Determines whether the surgery formula works!



Let $F_K = \sum_{n=0}^{\infty} f_n(q) x^n$ and $\delta_n(K) := \min \deg_q f_n(q)$

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Determines whether the surgery formula works!

Understanding the choice will help us solve the 

Non-perturbative complex Chern-Simons theory

$q \rightarrow w_p$: colored Jones, ADO, Alexander...

$$F_{\text{trefoil}}(x, q) = q x^2 + 7q x^3 + (7 + 28q - q^2) x^4 + (q^{-2} + 22q^{-1} + 48 + 89q - 7q^2 - 6q^3) x^5 + \dots$$

Non-perturbative complex Chern-Simons theory

$q \rightarrow w_p$: colored Jones, ADO, Alexander...

$$F_{\text{trefoil}}(x, q) = \boxed{q} \boxed{x^2} + \boxed{7q} x^3 + \boxed{(7 + 28q - q^2)} x^4 + \boxed{(q^{-2} + 22q^{-1} + 48 + 89q - 7q^2 - 6q^3)} x^5 + \dots$$

Homological grading in
knot Floer homology

3d-genus

Boundary slopes of essential surfaces

Non-perturbative complex Chern-Simons theory

$q \rightarrow w_p$: colored Jones, ADO, Alexander...

$$F_{\text{Knot}}(x, q) = \boxed{q} \boxed{x^2} + \boxed{7q} x^3 + \boxed{7} + \boxed{28q - q^2} x^4 + \boxed{q^{-2}} + \boxed{22q^{-1} + 48 + 89q - 7q^2 - 6q^3} x^5 + \dots$$

Diagram illustrating the relationship between the coefficients of the knot Floer homology generating function and various topological invariants:

- Homological grading in knot Floer homology:** Indicated by an orange arrow pointing to the coefficient q in the x^2 term.
- 3d-genus:** Indicated by a purple arrow pointing to the coefficient x^2 .
- Boundary slopes of essential surfaces:** Indicated by blue arrows pointing to the coefficients $7q$, 7 , q^{-2} , and the polynomial term in the x^5 term.
- Non-perturbative complex Chern-Simons theory:** Indicated by a green arrow pointing to the coefficient $28q - q^2$ in the x^4 term.
- Colored Jones, ADO, Alexander...:** Indicated by red boxes and red "???" labels above the x^4 and x^5 terms.



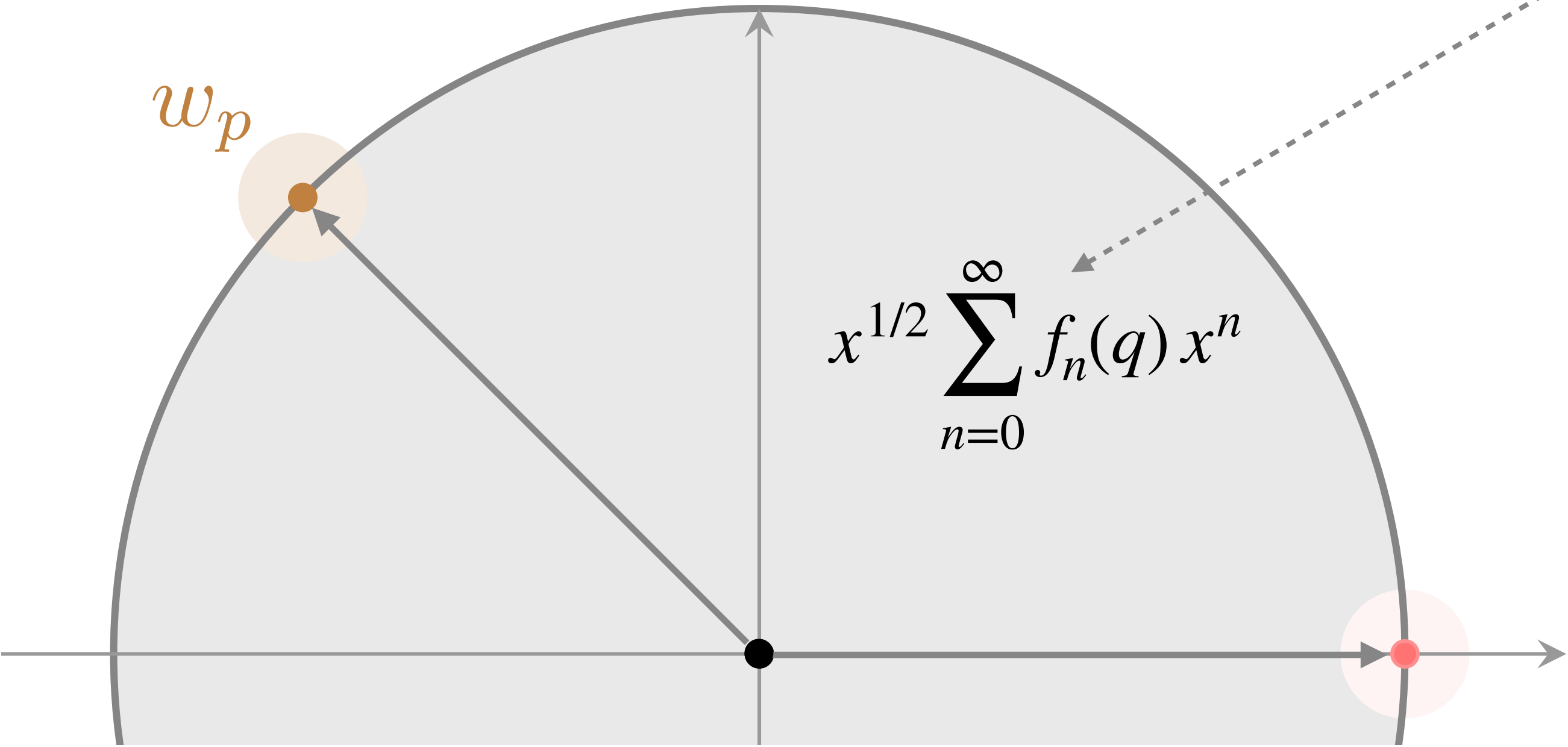
<https://topology.fyi>

OSSS., 2025

Gukov-Manolescu, 2018: resurgent analysis at $q = 1$

S. Park, 2020–21: quantum groups at generic q

$$\frac{ADO'_p}{\Delta_K(x^p)} + \sum_{j=1}^{\infty} \hbar^j_p \frac{P_{K,p}^{(j)}(x)}{\Delta_K^{2j+1}(x^p)}$$



$$\frac{x^{1/2} - x^{-1/2}}{\Delta_K(x)} + \sum_{j=1}^{\infty} \hbar^j \frac{P_K^{(j)}(x)}{\Delta_K^{2j+1}(x)} \quad q = 1 + \hbar$$

Alexander polynomial: $\Delta(\nearrow \searrow) - \Delta(\nwarrow \swarrow) = \left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right) \Delta(\searrow \swarrow) \quad \Delta(\bigcirc) = 1$